

Hydrocarbon Exploration and Licensing Policy (HELP)

Hydrocarbon Exploration and Licensing Policy (HELP) is a policy adopted by Government of India on 10.03.2016 indicating the new contractual and fiscal model for award of hydrocarbon acreages towards exploration and production (E&P). HELP is applicable for all future contracts to be awarded.

HELP replaces the present policy regime for exploration and production of oil and gas, known as New Exploration Licensing Policy (NELP), which has been in existence for 18 years.

Features of HELP

Four main aspects of HELP are:

- **Uniform License:** It provides for a uniform licensing system to cover all hydrocarbons such as oil, gas, coal bed methane etc. under a single licensing framework, instead of the present system of issuing separate licenses for each kind of hydrocarbons.
- **Open Acreages:** It gives the option to a hydrocarbon company to select the exploration blocks throughout the year without waiting for the formal bid round from the Government.
- **Revenue Sharing Model:** Present fiscal system of production sharing contract (PSC) is replaced by an easy to administer "revenue sharing model". The earlier contracts were based on the concept of profit sharing where profits are shared between Government and the contractor after recovery of cost. Under the profit sharing methodology, it became necessary for the Government to scrutinize cost details of private participants and this led to many delays and disputes. Under the new regime, the Government will not be concerned with the cost incurred and will receive a share of the gross revenue from the sale of oil, gas etc. Bidders will be required to quote revenue share in their bids and this will be a key parameter for selecting the winning bid. They will quote a different share at two levels of revenue called "lower revenue point" and "higher revenue point". Revenue share for intermediate points will be calculated by linear interpolation. The bidder giving the highest net present value of revenue share to the Government, as per transparent methodology, will get the maximum marks under this parameter.
- **Marketing and Pricing Freedom** has been granted, subject to a ceiling price limit, for new gas production from Deepwater, Ultra Deepwater and High Pressure-High Temperature Areas. The policy provides marketing and pricing freedom to the gas production from

existing discoveries which are yet to commence commercial production as on 1.1.2016 as well as for future discoveries^[1]. Considering the imperfections in gas markets in India, and to protect the interests of the consuming sector, a ceiling based on the landed cost of the alternate fuels has been imposed. The ceiling price shall be the, lowest of the

- › Fuel oil import landed price
- › Weighted average import landed price of substitute fuels (0.3 x price of imported coal + 0.4 x price of imported fuel oil + 0.3 x price of imported naphtha) and
- › LNG import landed price.

The ceiling will be calculated once in six months. The price data used shall be the trailing four quarters data with one quarter lag. To safeguard the Government revenue, the Government's share of profit will be calculated based on the higher of prevailing international crude price or actual price. All gas fields currently under production will continue to be governed by the pricing regime which is currently applicable to them.

Other features of HELP are:

- › Exploration is allowed through-out the contract period.
- › Exploration Phase for onshore areas have been increased from 7 years to 8 years and for offshore increased from 8 years to 10 years.
- › A concessional royalty regime will be implemented for deep water and ultra-deep water areas. These areas would not have any royalty for the first seven years (instead of the 5% at present), and thereafter would have a concessional royalty of 5% (in deep water areas) and 2% (in ultra-deep water areas), instead of the 10% at present. In shallow water areas, the royalty rates are reduced from 10% to 7.5%. For onshore areas royalty has been kept same i.e. 12.5% for oil and 10% for gas so that there is no impact on revenue to the State Governments.
- › This policy provides for a uniform, non-discretionary framework for extension of contract in respect of 28 Pre-NELP discovered fields. The extension will be granted for a period of 10 years both for oil and gas. During the extension period, it is proposed to increase the Government take by way of
 - › charging normal royalty and cess in place of concessional royalty and cess charged during the original contract period.
 - › The profit petroleum during extension period will also be 10 percent higher than the normal percentage

Objectives of HELP

The major Guiding Principles behind HELP are to:

- enhance domestic oil and gas production
- bring substantial investment
- generate sizable employment
- enhance transparency and
- reduce administrative discretion

Background

Till the adoption of Liberalisation policy in 1991-92, petroleum exploration and production (E&P) activities were carried out in India only by public sector oil companies viz, Oil and Natural Gas Corporation Limited (ONGC) and Oil India Limited (OIL). The New Exploration Licensing Policy (NELP) for exploration & production of oil & natural gas (but excluding Coal Bed Methane), and the Coal Bed Methane (CBM) Policy were formulated during 1997-98 by the Government of India, with Directorate General of Hydrocarbons (DGH) as the nodal agency, to provide a level playing field for both the public and private sector companies in exploration and production (E&P) of hydrocarbons. The activities in E&P sector have been significantly boosted by this policy and it has opened up E&P sector to private and foreign investment with 100% Foreign Direct Investment (FDI), bringing in a healthy competition between public sector oil companies and private sector or foreign companies.

Under NELP, which became effective in February 1999 (with the first production sharing contract (PSC) getting signed in 2000), acreages are offered to the participating companies through a process of open international competitive bidding, in a transparent manner with attractive terms & conditions. The first round of offer of blocks was launched in 1999 and most of the ninth round awards were concluded in 2012.

The salient features of NELP are as under:

- 100% FDI is allowed under NELP
- No mandatory state participation through ONGC/OIL or any carried interest of the Government.
- Blocks to be awarded through open international competitive bidding
- ONGC and OIL to compete for obtaining the petroleum exploration licenses (PEL) on a competitive basis instead of the existing system of granting them PELs on nomination basis.
- ONGC and OIL to get the same fiscal and contract terms as private companies.

- Freedom to the contractors for marketing of crude oil and gas in the domestic market.
- Royalty at the rate of 12.5% for the onland areas and 10% for offshore areas.
- Royalty to be charged at half the prevailing rate for deep water areas beyond 400 m bathymetry for the first 7 years after commencement of commercial production.
- Cess to be exempted for production from blocks offered under NELP.
- Companies to be exempted from payments of import duty on goods imported for petroleum operations.
- No signature, discovery or production bonuses.
- Agreement between government and contractor is governed by a Production Sharing Contract. A Model Production Sharing Contract is created which is reviewed for every NELP round.
- Contracts to be governed in accordance with applicable Indian Laws.

As at the end of nine rounds, 360 exploration blocks have been offered under NELP, and for 254 blocks PSCs have been signed. Presently, 166 blocks are active and 88 have been relinquished^[2]. Separately, under the CBM Policy-1997, thirty- four blocks have been offered and 33 were awarded as on date.

NELP Round & Launch year	No. of Blocks Offered	No of blocks bid for	No. of bids received	No. of B
I - 1999	48	28	45	25
II - 2000	25	23	44	23
III - 2002	27	24	52	23
IV - 2003	24	21	44	21
V - 2005	20	20	69	20
VI - 2006	55	52	165	52
VII - 2007	57	45	181	44
VIII - 2009	70	36	76	34

NELP Round & Launch year	No. of Blocks Offered	No of blocks bid for	No. of bids received	No. of B
IX - 2010	34	33	74	19
X - 2014	46			
TOTAL (excluding Xth round)	360	282	750	261

Source: DGH

The NELP bidding rounds have attracted many private and foreign companies, in addition to public sector oil companies. Before NELP, a total of 35 E&P Companies (5 PSUs, 15 Private sector Indian companies and 15 foreign companies) were working in India in Nomination blocks, Producing Fields and Pre-NELP blocks, either as Operator or Non-Operator^[3]. After conclusion of nine rounds of NELP, the number of companies increased to 117. This includes 11 PSUs, 58 private Indian companies and 48 foreign companies^[4]. A performance analysis of NELP may be seen from the Annual Reports of Director General of Hydrocarbons and from the website of Ministry of Petroleum .

However, NELP had many problems. Presently, there are separate policies and licenses for different hydrocarbons. There are separate policy regimes for conventional oil and gas, coal-bed methane, shale oil and gas and gas hydrates. Different fiscal terms are also in force for allocation of acreages for exploration for different hydrocarbons. In practice, there is overlapping of resources between different contracts. Unconventional hydrocarbons (shale gas and shale oil)^[5] were unknown when NELP was framed. This fragmented policy framework leads to inefficiencies in exploiting natural resources. For example, while exploring for one type of hydrocarbon, if a different one is found, it will need separate licensing, adding to cost.

The Production Sharing Contracts (PSCs) under NELP are based on the principle of "profit sharing". When a contractor discovers oil or gas, he is expected to share with the Government the profit from his venture, as per the percentage given in his bid. Until a profit is made, no share is given to Government, other than royalties and cesses. Since the contract requires the profit to be measured, it becomes necessary for the cost to be accounted for and checked by the Government. To prevent loss of Government revenue, these are requirements for Government approval at various stages to prevent the contractor from exaggerating the cost. Activities cannot be commenced till the approval is given. This process of approval of activities and cost gives the Government a lot of discretion and has become a major source of delays and disputes. Many projects have been delayed for months

and years due to disagreement between the Government and the contractor regarding the necessity or lack of necessity for particular items of cost, and the correctness of the cost.

Another feature of the current system is that exploration is confined to blocks which have been put on tender by the Government. There are situations where exploration companies may themselves have information or interest regarding other areas where they may like to pursue exploration. Currently these opportunities remain untapped, until and unless Government brings them to bidding at some stage.

The pricing of gas in the current system has undergone many changes and witnessed considerable litigation. Currently, the producer price of gas is fixed administratively by the Government. This has led to loss of revenue, a large number of disputes, arbitrations and court cases.

The current policy regime, in fixing royalties, does not distinguish between shallow water fields (where costs and risks are lower) and deep/ultra-deep water fields, where risks and costs are much higher.

The country faces a situation where oil and gas constitutes a major and increasing share of total imports. Oil production has stagnated while gas production has declined. There is a need for concerted policy measures to stimulate domestic production. Keeping in view this objective, the Government enunciated a new policy regime for exploration licensing, the Hydrocarbon Exploration and Licensing Policy, HELP

Source: Ministry of Petroleum and Natural Gas , accessed on 11 March 2016

Notes: Here, CBM stands for Coal Bed Methane and PEL and ML stands for Petroleum Exploration Licence (PEL) and Petroleum Mining Lease (PML): PEL is granted for a period of 7 years in onland /shallow water areas and for 8 years in deepwater and frontier areas for exploration activities as per PSC provisions under NELP. (This has been increased to 8 and 10 years respectively in the new fiscal regime adopted on 10.03.2016 –

HELP or Hydrocarbon exploration and Licensing Policy ↗) Petroleum Mining Lease (PML) is normally awarded for 20 years for producing Hydrocarbons as per The Oilfields Regulation & Development Act, 1948 and Petroleum & Natural Gas Rules, 1959. PEL/PML for offshore exploration & production operations is granted by the Union Government. In case of onland blocks, PEL/PML is granted by the concerned State Government on the basis of recommendation made by the Union Government for the awarded blocks.

A comparison of both the policies – HELP and NELP is given below:

Parameter	HELP	NELP
Fiscal Model	Revenue sharing	Profit sharing
Cost recovery	Not applicable	Yes
Cost efficiency	Encouraged	Neutral
Royalty	Low rates for offshore	Standard rates
Exploration Period	Onland and Shallow Water - 8 years Deepwater - 10 years	Onland and Shallow Water - 5 years Deepwater & Ultra-deepwater - 10 years
Management Committee	More focus on reservoir monitoring; no micro-management	Technical & financials
Revenue to Government	On production	After cost recovery i.e. net profit
Exploration in Mining Lease areas	Allowed	Not allowed
E&P activity for all hydrocarbons	Allowed	Not allowed

1. In the case of existing discoveries which are yet to commence commercial production as on 1.1.2016, if there is pending arbitration or litigation filed by the contractors directly pertaining to gas pricing covering such fields, this policy guideline shall apply only on the conclusion/withdrawal of such litigation/arbitration and the attendant legal proceedings.
2. Source: DGH; <http://www.dghindia.org/admin/Document/Topstory/13.pdf>
3. Petroleum Exploration Licenses (PEL) for domestic exploration & production of crude oil and natural gas were granted under four different regimes over a period of time:
 - a) Nomination Basis: Petroleum Exploration License (PEL) was granted to National Oil Companies viz. Oil and Natural Gas Corporation Ltd (ONGC) and Oil India Ltd. (OIL) on Nomination basis prior to implementation of NELP.
 - b) Pre-NELP Discovered Field: Petroleum Mining Lease (PML) was granted under small / medium size discovered field Production Sharing Contract (PSCs) during 1991 to 1993

where operators of blocks were private companies and ONGC/OIL has the participating interest.

- c) Pre-NELP Exploration Blocks: 28 Exploration Blocks were awarded to private companies between 1990 and prior to implementation of NELP where ONGC and OIL have the rights for participation in the block after hydrocarbon discoveries.
- d) New Exploration Licensing Policy (NELP) -1999 onwards: Under NELP, exploration blocks were awarded to Indian Private and foreign companies through international competitive bidding process where National Oil Companies viz, ONGC and OIL are also competing on equal footing.

Government of India has signed production sharing contracts for 28 discovered blocks, 28 exploration blocks under pre-NELP regime and 254 blocks under NELP regime with National Oil Companies and private (both Indian and foreign)/ Joint Venture companies as licensee for blocks. At present out of, 310 exploration blocks awarded so far under various bidding rounds (Discovered Field, Pre-NELP & NELP), 135 blocks/fields are operational. 17 blocks under nomination are being operated by ONGC and OIL.

4. Source: DGH; <http://www.dghindia.org/admin/Document/Topstory/13.pdf>

5. Unconventional Hydrocarbons are Coal bed methane, Gas Hydrates, Oil sands, shale oil etc. **Coal bed Methane (CBM)**, is an eco-friendly natural gas, stored in coal seams, generated during the process of the coalification (the degree of change undergone by coal as it matures from peat to anthracite). CBM exploration and exploitation has an important bearing on reducing the green house effect and earning carbon credit by preventing the direct emission of methane gas from operating mines to the atmosphere. Further, extraction of the CBM through degassing of the coal seams prior to mining of coal is a cost effective means of boosting coal production and maintaining safe methane level in working mines.

Gas hydrates are naturally occurring, crystalline, ice-like substances composed of gas molecules (methane, ethane, propane, etc.) held in a cage-like ice structure. (clathrate). Hydrates are a concentrated form of natural gas compared with compressed gas, but less concentrated than liquefied natural gas. It is estimated that a significant part of the Earth's fossil fuel is stored as gas hydrates, but as yet there is no agreement as to how large these reserves are. They are found abundantly worldwide in the top few hundred meters of sediment beneath continental margins at water depths between a few hundred and a few thousand feet and mainly in permafrost areas.

Oil sands or Tar Sands refers to crude trapped in sands in a semi solid form, mixed with sand and water. Tar Sands contain bitumen - a kind of heavy crude oil. They are found in Canada and Venezuela.

shale Oil is found in shale source rock that has not been exposed to heat or pressure long enough to convert trapped hydrocarbons into crude oil.

Oil Shales are usually fine-grained sedimentary rocks containing relatively large amounts of organic matter from which significant quantities of shale oil and combustible gas can be extracted by destructive distillation. The product thus generated is known as synthetic crude or more simply, syncrude. Oil shales are not technically shales and do not really contain oil. They are relatively hard rocks called marls - composed primarily of clay and calcium carbonate- containing a waxy substance called kerogen. The trapped kerogen can be converted into crude oil using heat and pressure to simulate natural processes. Included in most definitions of oil shale, either stated or implied, is the potential for the profitable extraction of shale oil and combustible gas or for burning as a fuel.

Tight Oil: Although the terms shale oil and tight oil are often used interchangeably in public discourse, shale formations are only a subset of all low permeability tight formations, which include sandstones and carbonates, as well as shales, as sources of tight oil production. Within the United States, the oil and natural gas industry typically refers to tight oil production rather than shale oil production, because it is a more encompassing and accurate term with respect to the geologic formations producing oil at any particular well.

Also see

- Revenue Sharing Contract (RSC)
- Open Acreage License Policy (OALP)
- Production Sharing Contract (PSC)
- New Exploration Licensing Policy (NELP)

References

1. *Ministry of Petroleum brief on Hydrocarbon exploration and Licensing Policy* [↗](#) issued on 10 March 2016
2. Press release [↗](#) of Ministry of Petroleum dated 10.03.2016
3. NELP brief by Ministry of Petroleum

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Ministry of Petroleum & Natural Gas



HELP Reforms Modernise India's Upstream Sector

Posted On: 11 DEC 2025 6:58PM by PIB Delhi

The Oilfield (Regulation and Development) Amendment Act, 2025 (ORD Act) was enacted and came into effect in 2025. The Amended ORD Act aims to foster an investor-friendly environment that enhances Ease of Doing Business (EoDB). The Petroleum and Natural Gas Rules, accordingly reflect the provisions of the Amended ORD Act.

Government had constituted a Joint Working Group (JWG), comprising representatives from key Exploration & Production (E&P) operators and Government, to assess and address issues affecting the EoDB in the upstream Oil and Gas sector. Pursuant to that, Government approved a set of reforms which *inter alia* include delivery point(s) within and outside the contract area, transfer of Participating Interest (PI) among existing PI holders under a contract and field handover process in DSF contracts etc.

Government have launched Hydrocarbon Exploration and Licensing Policy (HELP) in 2016. Under this policy, Open Acreage Licensing Policy (OALP) was started. A total of 172 exploration blocks covering 3,78,652 Square Kilometer have been awarded to successful bidders under 9 bidding rounds. Further, OALP bid Round X offering 25 explorations blocks covering 1,91,986.21 sq. km has been launched. It is the largest bid round so far in terms of acreage offered in a single OALP bid round under the HELP regime.

The salient features of HELP regime are as under:

- i. Reduced Royalty Rates,
- ii. No Oil Cess,
- iii. Uniform licensing system,
- iv. Revenue Sharing Model,
- v. Exploration rights on all retained area for full contract life,
- vi. Concessional Royalty Rates in case of early commercial production,
- vii. No Revenue Share based bidding in Blocks falling in Category-II and III Basins except in case of Wind fall gain etc.
- viii. Extended and phased exploration, bidding only 2D and 3D seismic and swapping of CWP with other surveys are key additional features for blocks falling in category II & III basins.
- ix. Originator Incentive Increased to 10 marks for Category-II & Category-III Basins

The measures for increasing the domestic exploration and production of hydrocarbon resources vis-à-vis the expansion of refining capacity and to reduce dependency on import of oil and gas *inter-alia* include:

- i. Policy for Relaxations, Extensions and Clarifications under Production Sharing Contract (PSC) regime for early monetization of hydrocarbon discoveries, 2014;
- ii. Discovered Small Field Policy, 2015;
- iii. Hydrocarbon Exploration and Licensing Policy (HELP), 2016;
- iv. Policy for Extension of PSCs, 2016 and 2017;
- v. Policy for early monetization of Coal Bed Methane (CBM), 2017;
- vi. Policy to Promote/Incentivize Enhanced Recovery Methods of Oil/Gas, 2018;
- vii. Policy Framework for exploration and exploitation of Unconventional Hydrocarbons including CBM, Shale Oil and Gas etc under Existing Contracts and Nomination Fields, 2018;
- viii. Release of about 1 million Sq. Km. (SKM) “No-Go” area in offshore in 2022 which were earlier blocked for exploration for decades.
- ix. The Oilfields (Regulation and Development) Amendment Act, 2025;

The policy reforms and incentives offered under HELP regime enhance the capability of the E&P industry to sustain and augment domestic production, thereby contributing to the strengthening of India’s long-term energy security.

This information was given by the MINISTER OF STATE IN THE MINISTRY OF PETROLEUM AND NATURAL GAS SHRI SURESH GOPI, in a written reply in Lok Sabha today.

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New Exploration and Licensing Policy (NELP)

New Exploration Licensing Policy (NELP) is a policy adopted by Government of India in 1997 indicating the new contractual and fiscal model for award of hydrocarbon acreages towards exploration and production (E&P). NELP was applicable for all contracts entered into by the Government between 1997 and 2016.

In March 2016, Hydrocarbon Exploration and Licensing Policy (HELP) replaced the extant policy regime for exploration and production of oil and gas -New Exploration Licensing Policy (NELP), which has been in existence for 18 years.

Objective

The main objective of NELP was to attract significant risk capital from Indian and Foreign companies, state of part technologies, new geological concepts and best management practices to explore oil and gas resources in the country to meet rising demands of oil and gas.

Background

Till the adoption of Liberalisation policy in 1991-92, petroleum exploration and production (E&P) activities were carried out in India only by public sector oil companies viz, Oil and Natural Gas Corporation Limited (ONGC) and Oil India Limited (OIL). NELP was formulated during 1997 by the Government of India, with Directorate General of Hydrocarbons (DGH) as the nodal agency, to provide a level playing field for both the public and private sector companies in exploration and production (E&P) of hydrocarbons. Since then, licenses for exploration are being awarded only through a competitive bidding system and National Oil Companies (NOCs) are required to compete on an equal footing with Indian and foreign companies to secure Petroleum Exploration Licences (PELs). The activities in E&P sector have been significantly boosted by this policy and it has opened up E&P sector to private and foreign investment with 100% Foreign Direct Investment (FDI). NELP was approved in 1997 and it became effective in February, 1999 (with the first production sharing contract (PSC) getting signed in 2000). The first round of offer of blocks was launched in 1999 and most of the ninth round awards were concluded in 2012. Nine rounds of bids have so far been concluded under NELP, in which production sharing contracts for 254 exploration blocks have been signed.

Features of NELP

The salient features of NELP are as under:

- 100% Foreign Direct Investment (FDI) is allowed under NELP
- No mandatory state participation through ONGC/OIL or any carried interest of the Government.
- Blocks to be awarded through open international competitive bidding
- ONGC and OIL to compete for obtaining the petroleum exploration licenses (PEL) on a competitive basis instead of the existing system of granting them PELs on nomination basis.
- ONGC and OIL to get the same fiscal and contract terms as private companies.
- Freedom to the contractors for marketing of crude oil and gas in the domestic market.
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- No signature, discovery or production bonuses.
- Agreement between government and contractor is governed by a Production Sharing Contract. A Model Production Sharing Contract is created which is reviewed for every NELP round.
- Contracts to be governed in accordance with applicable Indian Laws.

Performance of NELP

As at the end of nine rounds, 360 exploration blocks have been offered under NELP, and for 254 blocks, PSCs have been signed. Presently, 166 blocks are active and 88 have been relinquished^[1].

New Exploration Licensing Policy: Performance

NELP Round & Launch year	No. of Blocks Offered	No of blocks bid for	No. of bids received	No. of Blocks awarded
I- 1999	48	28	45	25
II-2000	25	23	44	23
III-2002	27	24	52	23
IV-2003	24	21	44	21
V-2005	20	20	69	20
VI-2006	55	52	165	52
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VIII-2009	70	36	76	34
IX-2010	34	33	74	19
X-2014	46			
TOTAL (excluding Xth round)	360	282	750	261

Source: DGH

The NELP bidding rounds have attracted many private and foreign companies, in addition to public sector oil companies. Before NELP, a total of 35 E&P Companies (5 PSUs, 15 Private sector Indian companies and 15 foreign companies) were working in India in Nomination blocks, Producing Fields and Pre-NELP blocks, either as Operator or Non-Operator^[2]. After conclusion of nine rounds of NELP, the number of companies increased to 117. This includes 11 PSUs, 58 private Indian companies and 48 foreign companies^[3]. A performance analysis of NELP may be seen from the Annual Reports of Director General of Hydrocarbons and from the website of Ministry of Petroleum .

However, NELP had many problems. Presently, there are separate policies and licenses for different hydrocarbons. There are separate policy regimes for conventional oil and gas, coal-bed methane, shale oil and gas and gas hydrates. Different fiscal terms are also in force for allocation of acreages for exploration for different hydrocarbons. In practice, there is overlapping of resources between different contracts. Unconventional hydrocarbons (shale gas and shale oil)^[4] were unknown when NELP was framed. This fragmented policy framework leads to inefficiencies in exploiting natural resources. For example, while exploring for one type of hydrocarbon, if a different one is found, it will need separate licensing, adding to cost.

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Another feature of the current system is that exploration is confined to blocks which have been put on tender by the Government. There are situations where exploration companies may themselves have information or interest regarding other areas where they may like to pursue for exploration. Currently these opportunities remain untapped, until and unless Government brings them to bidding at some stage.

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A comparison of both the policies – HELP and NELP is given below:

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Fiscal Model	Revenue sharing	Profit sharing
Cost recovery	Not applicable	Yes
Cost efficiency	Encouraged	Neutral
Royalty	Low rates for offshore	Standard rates
Exploration Period	Onland and Shallow Water - 8 years Deepwater - 10 years	Onland and Shallow Water - 7 ye Deepwater & Ultra-deepwater - 10
Management Committee	More focus on reservoir monitoring; no micro-management	Technical & financials examination
Revenue to Government	On production	After cost recovery i.e. from profit
Exploration in Mining Lease areas	Allowed	Not allowed

Parameter	HELP	NELP
E&P activity for all hydrocarbons	Allowed	Not allowed

- Source: DGH; <http://www.dghindia.org/admin/Document/Topstory/13.pdf>
- Petroleum Exploration Licenses (PEL) for domestic exploration & production of crude oil and natural gas were granted under four different regimes over a period of time:
 - Nomination Basis: Petroleum Exploration License (PEL) was granted to National Oil Companies viz. Oil and Natural Gas Corporation Ltd (ONGC) and Oil India Ltd. (OIL) on Nomination basis prior to implementation of NELP.
 - Pre-NELP Discovered Field: Petroleum Mining Lease (PML) was granted under small / medium size discovered field Production Sharing Contract (PSCs) during 1991 to 1993 where operators of blocks were private companies and ONGC/OIL has the participating interest.
 - Pre-NELP Exploration Blocks: 28 Exploration Blocks were awarded to private companies between 1990 and prior to implementation of NELP where ONGC and OIL have the rights for participation in the block after hydrocarbon discoveries.
 - New Exploration Licensing Policy (NELP) -1999 onwards: Under NELP, exploration blocks were awarded to Indian Private and foreign companies through international competitive bidding process where National Oil Companies viz, ONGC and OIL are also competing on equal footing.

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- Unconventional Hydrocarbons are Coal bed methane, Gas Hydrates, Oil sands, shale oil etc. **Coal bed Methane (CBM)**, is an eco-friendly natural gas, stored in coal seams, generated during the process of the coalification (the degree of change undergone by coal as it matures from peat to anthracite). CBM exploration and exploitation has an important bearing on reducing the green house effect and earning carbon credit by preventing the direct emission of methane gas from operating mines to the atmosphere. Further, extraction of the CBM through degassing of the coal seams prior to mining of

coal is a cost effective means of boosting coal production and maintaining safe methane level in working mines.

Gas hydrates are naturally occurring, crystalline, ice-like substances composed of gas molecules (methane, ethane, propane, etc.) held in a cage-like ice structure. (clathrate). Hydrates are a concentrated form of natural gas compared with compressed gas, but less concentrated than liquefied natural gas. It is estimated that a significant part of the Earth's fossil fuel is stored as gas hydrates, but as yet there is no agreement as to how large these reserves are. They are found abundantly worldwide in the top few hundred meters of sediment beneath continental margins at water depths between a few hundred and a few thousand feet and mainly in permafrost areas.

Oil sands or Tar Sands refers to crude trapped in sands in a semi solid form, mixed with sand and water. Tar Sands contain bitumen - a kind of heavy crude oil. They are found in Canada and Venezuela.

shale Oil is found in shale source rock that has not been exposed to heat or pressure long enough to convert trapped hydrocarbons into crude oil.

Oil Shales are usually fine-grained sedimentary rocks containing relatively large amounts of organic matter from which significant quantities of shale oil and combustible gas can be extracted by destructive distillation. The product thus generated is known as synthetic crude or more simply, syncrude. Oil shales are not technically shales and do not really contain oil. They are relatively hard rocks called marls - composed primarily of clay and calcium carbonate- containing a waxy substance called kerogen. The trapped kerogen can be converted into crude oil using heat and pressure to simulate natural processes. Included in most definitions of oil shale, either stated or implied, is the potential for the profitable extraction of shale oil and combustible gas or for burning as a fuel.

Tight Oil: Although the terms shale oil and tight oil are often used interchangeably in public discourse, shale formations are only a subset of all low permeability tight formations, which include sandstones and carbonates, as well as shales, as sources of tight oil production. Within the United States, the oil and natural gas industry typically refers to tight oil production rather than shale oil production, because it is a more encompassing and accurate term with respect to the geologic formations producing oil at any particular well.

Also See

- Hydrocarbon Exploration and Licensing Policy (HELP)
- Revenue Sharing Contract (RSC)

- Open Acreage License Policy (OALP)
- Production Sharing Contract (PSC)

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Environmental Management in India's Upstream Oil and Gas Sector: A Comprehensive Impact Assessment

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Abstract:

The upstream oil and gas sector in India has seen substantial growth in recent years, raising concerns about its environmental impact. The expansion of oil and gas operations has led to environmental degradation, including air and water pollution, soil contamination, and disruption of local ecosystems. Understanding the extent of these environmental challenges is crucial to assess the damage caused by these activities. While there are existing environmental laws, regulations, and policies for the upstream sector, ensuring compliance remains critical. This study aims to identify gaps or areas of non-compliance in environmental management. Additionally, the adoption and effectiveness of various mitigation measures and technologies will be explored to reduce the environmental footprint of these operations. As India seeks to balance economic growth with environmental sustainability, this research will examine the long-term impact of environmental management practices and their contribution to a sustainable future.

Keywords: Upstream Oil and Gas, Environmental Impact, Mitigation Measures, Sustainability

1. Introduction:

India's upstream oil and gas sector plays a pivotal role in the nation's energy security and economic development. However, the sector's rapid growth has raised significant concerns regarding its environmental impact. Exploration and extraction activities in this sector have the potential to cause severe environmental degradation, including air and water pollution, soil contamination, and biodiversity loss. As India seeks to address its growing energy needs while managing environmental challenges, there is a pressing need to implement effective environmental management practices that balance resource extraction with ecological preservation.

The environmental implications of oil and gas exploration are multifaceted, encompassing impacts on air quality, water bodies, soil health, and surrounding ecosystems. The drilling process, in particular, generates large volumes of waste, such as drill cuttings, drilling fluids, and chemical residues, which are often stored in surface pits or tanks before disposal. Improper disposal of these byproducts can lead to long-term contamination of the environment. Therefore, robust environmental regulations are crucial to ensure that waste is managed appropriately, and that wells are properly decommissioned once they reach the end of their operational life.

This analysis aims to evaluate the regulatory frameworks governing India's upstream oil and gas sector, focusing on the industry's approach to addressing environmental concerns. The study will also examine the sector's adherence to environmental laws and regulations, comparing these practices with international standards to identify potential areas for improvement.

Beyond regulatory compliance, the study emphasizes that effective environmental management practices can act as a catalyst for sustainable development in the sector. With growing global awareness of climate change and the shift towards cleaner energy sources, India's upstream oil and gas sector is at a critical juncture. This research seeks to provide insights into how the sector can enhance its environmental practices, contributing to both energy security and sustainable development.

Through a comprehensive review, the study aims to offer recommendations that could guide the sector towards more responsible and sustainable practices. It is proposed that by advancing environmental management, India's upstream oil and gas industry can play a crucial role in the nation's journey towards sustainable development.

2. Legal and Regulatory Framework Governing India's Upstream Oil and Gas Sector

The Indian government has periodically introduced various licensing frameworks to enhance domestic oil and gas production. Blocks awarded under specific regimes are governed by the regulations of that regime, even if amendments

are made to the regime afterward. Currently, the Government of India has entered into 310 production-sharing contracts for 29 discovered fields, 28 pre-NELP exploration blocks, and 254 NELP regime blocks¹.

Legal and Regulatory Structure

India operates under a federal constitution, where legislative powers are shared between central and state legislatures. According to Article 246 of the Indian Constitution, the Union Parliament holds authority over regulating and developing oil fields, mineral oil resources, petroleum, and petroleum products. Meanwhile, state governments have jurisdiction over issues such as land use, labour, water, and local governance. Therefore, while exploration and production contracts for hydrocarbons are negotiated by the Indian government, state governments issue licenses and approvals for onshore activities. For offshore blocks, licensing authority lies with the Indian government.

A. Key Domestic Legislation Governing Oil and Gas Operations

Several key pieces of legislation regulate India's upstream oil and gas sector:

1. **Oilfields (Regulation and Development) Act, 1948 (Oilfields Act):** This Act governs the processes for licensing the extraction of petroleum, natural gas, crude oil, refined oil, and other petroleum products in both liquid and solid forms. It grants the Indian government rule-making power for mining leases and mineral oil development, as well as the authority to determine royalty rates.
2. **Petroleum and Natural Gas Rules, 1959 (PNG Rules):** The PNG Rules, enacted under the Oilfields Act, regulate mining leases and exploration licenses concerning petroleum and natural gas that belong to the Government of India. These rules aim to ensure the conservation and development of petroleum and natural gas resources. They also prohibit petroleum prospecting or mining without a Petroleum Exploration License (PEL) or Petroleum Mining Lease (PML), both granted under the PNG Rules².
3. **Mines Act, 1952, and Oil Mines Regulations, 2017:** The Mines Act provides provisions for the health, safety, and welfare of workers in the mining industry, including oil mines. It outlines the responsibilities of mine owners regarding operations and management, along with health and safety protocols. The Oil Mines Regulations set safety criteria for upstream oil and gas facilities, focusing on safety protocols and mandatory training. These regulations stipulate that all oil and gas production installations must have a flare line with specific requirements, such as the height of the flare stack and the installation of remote ignition systems.
4. **Petroleum and Natural Gas (Safety in Offshore Operations) Rules, 2008 (PNG Safety Rules):** Established in 2008, these regulations address safety concerns in offshore oil and gas exploration, extraction, and management. They cover aspects such as preparation of records, consent requirements, safety measures, environmental protection, health protocols, and penalties for non-compliance³.

In addition to these legislative provisions, the Indian government periodically releases policies, standards, directives, and guidelines to govern various facets of the upstream oil and gas sector, ensuring that operations align with national priorities and environmental responsibilities.

B. Regulatory Authorities

Several key regulatory and administrative bodies oversee the upstream oil and gas sector in India:

1. **Ministry of Petroleum and Natural Gas (MoPNG):** The MoPNG is the central authority responsible for overseeing the exploration and production of oil and natural gas, as well as the refining, distribution, and marketing of petroleum products. The Ministry also manages the import, export, and conservation of petroleum resources. Acknowledging the critical role of oil and gas in India's economy, the MoPNG has implemented various initiatives to boost domestic production and optimize petroleum resources, with a focus on priorities such as energy efficiency, sustainability, access, and security.

C. Regulatory Bodies and Legislation

Several key regulatory bodies oversee the upstream oil and gas sector in India, each playing a vital role in ensuring the industry's compliance with safety, environmental, and operational standards:

1. **Directorate General of Hydrocarbons (DGH):** Established in 1993 under the Ministry of Petroleum & Natural Gas, the DGH manages India's oil and natural gas resources with an emphasis on environmental, safety, technological, and economic considerations. The DGH is responsible for implementing the New Exploration Licensing Policy (NELP), overseeing Production Sharing Contracts (PSCs), encouraging investment in exploration and production, and exploring alternative hydrocarbon resources such as Coal Bed Methane, Gas Hydrates, and Oil Shales.

¹ Ministry of Petroleum and Natural Gas (MoPNG), Government of India. (2023). *Annual Report 2022-2023*.

² Government of India. (1959). *Petroleum and Natural Gas Rules, 1959*.

³ Government of India. (2008). *Petroleum and Natural Gas (Safety in Offshore Operations) Rules, 2008*

2. **Oil Industry Safety Directorate (OISD):** Operating within the Ministry of Petroleum and Natural Gas, the OISD is a technical directorate focused on developing and enforcing self-regulatory safety measures to improve safety standards across the oil and gas industry in India.
3. **Directorate General of Mines Safety (DGMS):** The DGMS, under the Ministry of Labour and Employment, is responsible for overseeing the safety, health, and welfare of workers in mines, including oil and gas facilities. The agency ensures compliance with health and safety regulations to protect workers in hazardous environments.

Apart from these bodies, various laws and regulations address environmental protection, labour rights, and other aspects relevant to the oil and gas sector in India. Some key regulations include:

1. **Environmental Impact Assessment (EIA):** EIA is a process used to predict the potential environmental impacts of a proposed project. It helps decision-makers compare alternatives to identify the most balanced approach in terms of economic and environmental outcomes. The process systematically evaluates both positive and negative impacts and suggests mitigation measures. India's EIA process, which began in the 1970s with a focus on river valley projects, was formalized in 1994 under the Ministry of Environment and Forests (MEF). Since then, there have been several amendments to the EIA notification, expanding its scope to include various types of projects, with specific requirements for coastal activities and international funding⁴.
2. **Water (Prevention and Control of Pollution) Act, 1974 (WPCP):** This Act applies to several Indian states and aims to prevent water pollution and maintain water quality. It prohibits the release of pollutants into water bodies beyond specified limits and establishes penalties for violations. The Act is relevant to the oil and gas sector as it addresses potential impacts on water resources, requiring compliance with regulations designed to prevent contamination of water sources⁵.
3. **Air (Prevention and Control of Pollution) Act, 1981 (APCP):** The APCP Act sets standards for air quality and establishes mechanisms for controlling air pollution. It regulates the use of polluting fuels and substances and requires industries to obtain consent from state pollution control boards to operate in pollution-sensitive areas. The CPCB enforces national air quality standards, and the Act empowers pollution control boards to take emergency action in cases of non-compliance⁶.
4. **The Forest (Conservation) Act, 1980:** This Act aims to protect and conserve forests by regulating diversion of forest land for non-forestry purposes. It limits state authorities' power to de-reserve forests or divert forest land for activities such as agriculture or urban development. It required prior approval of government of India for diversion of forest land for non-forestry purposes.
5. **Wildlife Protection Act, 1972:** While this Act primarily focuses on the conservation of wildlife, it also applies to activities in the oil and gas sector that could affect wildlife habitats. Compliance with this Act is essential to ensure that oil and gas exploration and production activities do not harm wildlife or ecosystems.
6. **The Public Liability Insurance Act, 1991:** This Act requires industries handling hazardous substances to secure insurance coverage for liabilities arising from accidents, including loss of life, injury, or property damage. Although it is not specific to the oil and gas sector, companies involved in the storage and transportation of hazardous materials must comply with this requirement to manage potential public liabilities.
7. **National Green Tribunal (NGT) Act, 2010:** The NGT was established to ensure efficient resolution of environmental cases, including those related to the conservation of forests, wildlife, and natural resources. It provides a platform for individuals and communities to seek relief and compensation for environmental damage and enforces legal rights related to environmental protection.
8. **Biological Diversity Act, 2002:** This Act focuses on conserving biological diversity, ensuring sustainable use of resources, and preventing biopiracy. It mandates benefit-sharing from biological resources and their derivatives, promoting equitable distribution of benefits among local communities and stakeholders.
9. **Environmental (Protection) Act, 1986:** The Environmental Protection Act provides a comprehensive framework for environmental management in India. It empowers the central government to regulate industries, set emission standards, and manage hazardous waste. The Act mandates Environmental Impact Assessments (EIAs) for specific projects and includes provisions for controlling pollution in sensitive areas, such as coastal zones and ecologically fragile regions.

⁴ Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India. (2006). *Environmental Impact Assessment Notification, 2006*

⁵ Government of India. (1974). *Water (Prevention and Control of Pollution) Act, 1974*. Ministry of Environment, Forest and Climate Change, Government of India

⁶ Government of India. (1981). *Air (Prevention and Control of Pollution) Act, 1981*. Ministry of Environment, Forest and Climate Change, Government of India. Retrieved from <http://legislative.gov.in/sites/default/files/A1981-14.pdf>.

These regulations collectively govern the operations and environmental responsibilities of the oil and gas industry in India, ensuring that industry practices comply with safety, environmental, and social standards.

3.Environmental Management Practices

The upstream oil and gas industry faces a range of environmental challenges, including liquid, gaseous, and solid pollution. Addressing these challenges effectively requires proactive planning, appropriate budget allocation, and a commitment to minimizing environmental impacts. Air quality management begins with baseline assessments and dispersion modelling during facility design, ensuring that emission limits are met and accounted for in equipment selection. The practice of flaring and venting natural gas, which contributes to emissions and can negatively impact local communities, must adhere to international standards for reduction.

Produced water management is an essential aspect, with disposal options such as injection into reservoirs or dedicated wells carefully evaluated. The industry also emphasizes waste segregation, recycling, and proper disposal planning to address both hazardous and non-hazardous wastes. When handling drilling fluids and cuttings, various treatment and disposal methods are employed, ensuring compliance with environmental regulations. Noise management and decommissioning are considered, with a focus on restoring landforms and successfully implementing revegetation efforts. The entire environmental management process reflects the industry's commitment to sustainable practices and environmental stewardship. In India, exploration and production (E&P) projects must comply with national and state regulations, including obtaining clearances such as Environmental Clearance (EC), Forest Clearance (FC), Consent to Operate (CO), Consent to Establish (CE), and Hazardous Waste Authorization. These processes involve thorough scrutiny, public consultation, and necessary approvals.

The oil and gas sector also pays water cess on a quarterly basis, submits compliance reports, and adheres to the conditions stipulated by regulatory authorities. The exploitation of oil reserves inevitably raises environmental concerns, highlighting the importance of effectively implementing statutory provisions. Understanding the relationship between oil field exploration, development, and environmental management is vital to ensuring that these activities align with sustainability goals⁷.

4.Industry Compliance with Environmental Legislation and Regulations

Compliance with environmental regulations in India's upstream oil and gas sector is a complex and evolving process shaped by several factors, including regulatory oversight, corporate responsibility, technological advancements, public and community pressures, adherence to international standards, and government policies. The industry's compliance level is influenced by the effectiveness of environmental impact assessments, air and water pollution control measures, and overall commitment to sustainable, eco-friendly practices. Ongoing monitoring, periodic audits, and transparent reporting are vital for maintaining and enhancing compliance standards across the sector.

Prominent Indian companies in the upstream oil and gas sector, such as Oil and Natural Gas Corporation Limited (ONGC), Oil India Limited (OIL), Reliance Industries Limited (RIL), GAIL (India) Limited, Indian Oil Corporation Limited (IOCL), and Hindustan Petroleum Corporation Limited (HPCL), are increasingly prioritizing environmental management and sustainability.

ONGC (Oil and Natural Gas Corporation Limited) focuses heavily on environmental responsibility in its Exploration and Production operations. The company integrates sustainability into its operations through investments in advanced technology, waste management, and biodiversity conservation. ONGC has made significant strides in reducing natural gas flaring, controlling noise pollution, treating produced water, and adopting green practices. It also actively participates in biodiversity conservation projects. The company's commitment to environmental protection is evident in its sustainable development policies, which emphasize proactive strategies, technology adoption, and community engagement. ONGC's achievements in environmental performance have earned it recognition through various awards⁸.

Oil India Limited (OIL) emphasizes environmental responsibility, particularly in energy, emissions, water, and waste management. The company has committed to reducing greenhouse gas emissions and addressing climate change through responsible energy usage and waste management practices. OIL incorporates renewable energy solutions, partners in carbon reduction projects, and closely monitors air and water quality, ensuring efficient resource use and sustainable waste management⁹.

⁷ ONGC. (2020). *Sustainability Report 2020*. Oil and Natural Gas Corporation

⁸ ONGC, <https://ongcindia.com/web/eng/sustainability/environment/environment-management>, (last visited on December 27th, 2023)

⁹ OIL, https://www.oil-india.com/Document/OIL_Sustainability_Report2021-22.pdf, (last visited on November 26th, 2023)

Reliance Industries Limited (RIL) integrates environmental sustainability into its Health, Safety, and Environment (HSE) vision, focusing on energy efficiency, waste reduction, and regulatory compliance. RIL follows a holistic approach that includes clean energy practices, biodiversity conservation, and digitization. The company tracks and manages its greenhouse gas emissions, invests in retrofit technologies, and ensures compliance through environmental audits and approvals. By embracing circular economy principles, RIL reduces, reuses, and recycles, particularly in hazardous waste management. Acknowledging the global shift towards a low-carbon economy, RIL aligns its strategies with investor expectations, strengthening internal governance, and investing in R&D for carbon reduction¹⁰.

Indian Oil Corporation Limited (Indian Oil) places a strong emphasis on Health, Safety, and Environment (HSE) as foundational principles for sustainable development. The company actively engages in pollution control, resource conservation, and renewable energy initiatives, demonstrating a deep commitment to environmental responsibility. Indian Oil prioritizes creating safe workplaces, promoting resource efficiency, and contributing to the well-being of its employees, customers, and communities.

In conclusion, while these companies exhibit a strong commitment to environmental sustainability and responsibility, a thorough evaluation is necessary to determine the true extent of their compliance with environmental regulations. Assessing factors such as regulatory adherence, transparency in reporting, and the tangible environmental impacts of their initiatives will provide a clearer understanding of their efforts. Ongoing monitoring, innovation, and community involvement will be key in ensuring that these companies continue to meet their environmental commitments and make meaningful contributions to global sustainability goals.

4. International Environmental Management Practices

A. Strategic Environmental Assessment (SEA)

SEA is a process that evaluates the environmental impacts of proposed policies, plans, or programs at an early stage, ensuring these impacts are considered alongside economic and social factors. It helps identify cumulative environmental effects from various development scenarios. While SEAs are valuable for large-scale planning, they are not always legally mandated in the oil and gas sector but are often utilized for major policy decisions. A notable example is Uganda's Albertine Graben oil discovery, where an SEA conducted in 2013 helped assess the environmental and social impacts of oil and gas activities, influencing sustainable management practices for the industry¹¹.

B. B. Environmental, Social, and Health Impact Assessments (ESHIA)

ESHIA evaluates the potential impacts of development projects, identifying risks and mitigation measures. It is commonly used in the oil and gas industry for facility construction, decommissioning, and reclamation activities. ESHIA is often required by host governments and is critical in ensuring compliance with environmental, social, and health regulations. It also provides input into SEAs, creating a comprehensive approach to managing environmental risks.

C. C. Environmental Management Systems (EMS) and Operating Management Systems (OMS)

EMS is a company-wide framework for managing environmental performance, ensuring compliance with regulations, and reducing environmental impacts. In the oil and gas sector, EMS works alongside OMS, which covers broader operational aspects like health, safety, and security. OMS helps integrate risk management practices, including environmental, social, and health concerns, with a focus on continuous improvement and adherence to legal and corporate standards.

D. D. Environmental, Social, and Health Management Plans (ESHMP)

An ESHMP is part of the OMS, detailing mitigation measures, compliance strategies, and performance optimization efforts. It integrates environmental management with other operational aspects and ensures commitments from ESHIA are effectively tracked and implemented. The ESHMP outlines the roles, responsibilities, and actions for managing environmental and social impacts, including waste management and community engagement, from the planning stage through to project completion.

E. E. Emergency Preparedness and Response

Oil and gas companies must prepare for incidents like oil spills or natural disasters. Emergency plans should be

¹⁰ Reliance Industries Limited,
<https://www.ril.com/Sustainability/HealthSafety.aspx#:~:text=We%20comply%20with%20all%20applicable.facility%20design%20and%20operating%20procedures.,> (last visited on November 26th, 2023)

¹¹ Saddle, B., & Verheem, R. (1996). *Environmental management in the 1990s: A social and political perspective*

integrated into the OMS and designed to minimize environmental impacts, ensure safety, and enable rapid response. Drawing from past incidents like the Deepwater Horizon spill, emergency plans are regularly updated, with ongoing training and exercises to maintain preparedness. Monitoring and reporting on incidents help refine response strategies and ensure regulatory compliance.

F. EU Offshore Safety Regulations

In the European Union, offshore oil and gas operations are regulated under the Directive on Safety of Offshore Oil and Gas Operations. This directive mandates risk assessments, safety measures, and financial liability for environmental damage. It was introduced following incidents like the Deepwater Horizon spill and emphasizes strict safety protocols. Companies are required to adopt these regulations, enhancing safety and reducing the risk of major environmental accidents.

5. Comparing International and Indian Environmental Management Practices

India's environmental management practices in the oil and gas sector are aligned with international standards, focusing on comprehensive assessments such as EIA and SEA. However, international practices tend to be more standardized and rigorous in their enforcement, with greater emphasis on global emission reduction, waste management, and adherence to detailed methodologies. While India is making progress, challenges remain in ensuring consistency in enforcement and adapting to the evolving demands of the oil and gas sector. International practices are more robust due to well-established regulatory structures, resources, and expertise, facilitating more effective monitoring, public participation, and transparency.

In conclusion, while India is progressing in aligning its practices with international standards, it faces challenges in enforcement consistency and adapting to the dynamic nature of the oil and gas industry. Addressing these challenges will be essential for creating a more sustainable and effective environmental management framework.

6. Sustainable Practices for Indian Upstream Oil and Gas Sector

To improve sustainability and energy efficiency in the upstream oil and gas sector, India is focusing on various strategies, including energy-saving initiatives, enhanced oil recovery techniques, and reduction of gas flaring. Key efforts involve adopting advanced technologies, optimizing energy use, and addressing environmental impacts. Notable companies like ONGC, OIL, and GAIL have taken significant steps toward energy conservation and emissions reduction, setting examples for the industry.

1. Energy Efficiency Strategies

Indian oil and gas companies are implementing various initiatives to enhance energy efficiency:

- **Adopting Energy Conservation Technologies:** Installation of energy-saving systems at different facilities.
- **Efficient Lighting Systems:** Use of energy-efficient lighting technologies.
- **Bi-Fuel Technology:** Incorporation of bi-fuel systems to reduce energy consumption.
- **Conducting Energy Audits:** Regular assessments to identify areas of energy wastage and potential savings.

For instance, ONGC focuses on optimizing systems that have resulted in substantial electricity savings, while OIL ensures energy self-sufficiency by pursuing various conservation measures. GAIL has set a target to reduce specific energy consumption by 5% through integrated energy management and renewable energy integration. In addition, regulatory bodies can further promote energy efficiency by offering incentives, mandates, and fostering awareness around best practices.

2. Reducing Gas Flaring

Gas flaring, which involves burning off excess natural gas during oil extraction, is a significant contributor to global CO₂ emissions. While flaring regulations exist in India, challenges remain regarding enforcement, infrastructure, and pricing, hindering effective commercialization of associated gas.

International organizations like the World Bank recommend clearer legislation, fiscal incentives, and stronger market frameworks to reduce flaring. In India, ONGC and companies in regions like West Bengal have made efforts to reduce flaring through technological solutions, with continued emphasis on achieving more efficient gas utilization. Globally, initiatives such as the Global Gas Flaring Reduction Partnership are driving efforts to curb these emissions¹².

3. Greenhouse Gas Reduction and Transition to Renewable Energy

To ensure long-term sustainability, India's upstream oil and gas sector must reduce greenhouse gas emissions and accelerate the adoption of renewable energy sources. In the short term, this includes mandatory disclosures of energy

¹² IPIECA, *Environmental management in the upstream oil and gas industry*, Report 254, 46 (2020), <https://www.ipieca.org/resources/environmental-management-in-the-upstream-oil-and-gas-industry>

usage and carbon emissions, incentivizing energy-saving initiatives in exploration and production, and offering tax incentives to reduce flaring.

Looking toward the medium term, it is recommended that the government enforce reductions in associated gas flaring in line with global standards and encourage investments in renewable energy through Clean Development Mechanism (CDM) projects. These projects could aim to generate renewable energy equal to or greater than the energy consumed in hydrocarbon extraction and processing¹³.

In the long term, a comprehensive infrastructure plan is needed to commercialize associated natural gas, which is currently flared due to a lack of utilization infrastructure.

4. Policy Framework for Sustainability

To support a greener and more sustainable future for India's upstream oil and gas sector, a well-defined public policy framework should be established. This framework should draw from both domestic achievements and international best practices, fostering an environment where sustainability initiatives are promoted, incentivized, and made feasible across the sector. By doing so, India can improve its energy efficiency, reduce environmental impacts, and move toward a low-carbon economy while recognizing the essential role of oil and gas in the country's economic landscape.

In conclusion, while the Indian upstream oil and gas industry has made significant strides in integrating sustainability practices, there are ample opportunities for further enhancement. By strengthening policy, investing in renewable energy, and reducing emissions, the sector can contribute to a more sustainable energy future.

4.1 Enhancing Sustainability: Environmental Management Approaches for India's Upstream Oil and Gas Sector

Integrating Strategic Environmental Assessment (SEA) into development plans, including poverty reduction strategies, significantly improves the effectiveness of these initiatives. For example, Tanzania's second poverty reduction strategy included SEA, which helped incorporate environmental, social, and economic factors into sector policies and national budgets. SEA is a vital tool for integrating sustainability into decision-making, offering confidence to policymakers while helping avoid unforeseen negative outcomes. It identifies opportunities for positive impacts and highlights the consequences of policy decisions, as evidenced by Ghana's National Strategy for Growth and Poverty Reduction and Pakistan's Thermal Power Generation Policy. SEA acts as a safeguard against costly errors in policymaking, helping prevent unexpected costs or the need for reparation. Furthermore, SEA encourages improved governance by involving the public in decision-making processes. A case in South Africa, for instance, demonstrates how public engagement in water use studies can resolve complex resource allocation challenges. SEA also promotes international cooperation by addressing shared environmental concerns, contributing to the achievement of MDG 7, which focuses on ensuring environmental sustainability. In essence, SEA plays a pivotal role in fostering sustainable development by minimizing negative environmental impacts and enhancing resource opportunities.

Environmental Impact Assessment (EIA) is a fundamental process in ensuring that major projects, especially in the oil and gas sector, do not compromise essential natural resources such as water, land, and biodiversity. Projects often lead to deforestation, land degradation, and air and water pollution, making it critical to strike a balance between environmental protection and development. EIA has three key functions: predicting potential issues, suggesting strategies to avoid harm, and enhancing positive outcomes. For the past 25 years, organizations like the International Institute for Environment and Development (IIED) have advocated for sustainable development, seeking to ensure that future generations enjoy a quality of life equal to or better than the present. Sustainable development requires trade-offs, often between economic growth and environmental conservation. EIA, through public consultation and stakeholder engagement, offers a structured approach to decision-making by identifying conflicts, predicting constraints, and recommending mitigation measures to optimize environmental outcomes. India's comprehensive EIA framework ensures the oil and gas sector adheres to environmental protection standards while fostering informed decision-making regarding potential environmental impacts. Environmental, Social, and Governance (ESG) practices are gaining traction in the oil and gas industry, driven by the growing demand to address climate change and reduce methane emissions. Major industry players like ExxonMobil, Total, and Shell are increasingly focusing on ESG goals, aligning with investor expectations and government regulations. The failure to adopt ESG principles can lead to reputational damage and reduced access to capital. Adopting ESG-focused engineering solutions, which emphasize cleaner, more cost-effective technologies, can result in substantial benefits such as improved ethical practices, cost savings, and enhanced capital attraction. ESG-friendly strategies not only promote environmental and social responsibility but also contribute to long-term profitability by helping companies navigate ethical dilemmas and support sustainable business practices.

In conclusion, the integration of robust environmental management practices—such as SEA, EIA, and ESG principles—into the development process is crucial for promoting sustainability within the oil and gas sector. These practices help

¹³ Saddler, B., & Verheem, R. (1996). *Greenhouse gas reduction and transition to renewable energy*

mitigate environmental impacts, foster better governance, and unlock opportunities for sustainable growth. Oil and gas companies that embrace environmental responsibility and cost-effective solutions will benefit from improved profitability and long-term viability.

7. Oil and Gas Companies: Contributors to Sustainability or Barriers?

The urgency of addressing climate change is escalating, driven by more frequent extreme weather events, alarming global reports, and increasing public protests. The world's largest greenhouse gas (GHG) emitters—such as China, Saudi Arabia, India, and Canada—are currently not on track to meet their Paris Agreement commitments, according to the Climate Action Tracker. As pressure mounts for the complete removal of fossil fuels from the energy system, oil and gas companies are facing heightened scrutiny from political and environmental leaders. The industry's challenges are compounded by policies designed to decarbonize, raising concerns among investors about the potential ceiling for future hydrocarbon demand¹⁴.

Despite these challenges, the oil and gas industry still hold a vital role in the global energy landscape, especially with the projected growth in global energy demand. To navigate the transition toward a low-carbon economy, oil and gas companies are diversifying their business models, investing in decarbonization technologies, reassessing geopolitical risks, and adopting climate-focused Environmental, Social, and Governance (ESG) principles. The industry's ability to transition from focusing on "oil and gas" to a broader "energy" approach offers an opportunity to mitigate risks associated with the changing energy landscape¹⁵.

The oil and gas sector faces a strategic dilemma of balancing short-term profits with long-term sustainability. Investments in clean energy technologies currently constitute a very small portion of overall capital expenditure, but the industry has the potential to make substantial strides today in reducing its environmental impact. Relying solely on electricity for the energy transformation is insufficient—oil and gas companies can play a key role in advancing capital-intensive clean energy technologies. While the transition to a cleaner energy sector may proceed without the direct involvement of oil and gas, it would be far more costly and challenging¹⁶.

In conclusion, while oil and gas companies face significant challenges in the face of climate urgency and pressure for decarbonization, they also have the potential to contribute significantly to the solution. By evolving their business models, supporting clean technologies, and embracing sustainability, these companies can play a pivotal role in the successful energy transition, despite the inherent difficulties.

8. Conclusion

Despite growing global momentum toward sustainable practices, current trajectories indicate that the world is not on track to meet the Sustainable Development Scenario (SDS), which envisions a significant reduction in oil and gas demand by 2040. This gap in emissions reduction, alongside rising public support for low-emission policies, has intensified the call for decisive action. The SDS provides a roadmap for the energy sector's rapid transition, addressing not only emissions but broader imperatives like universal energy access, cleaner air, and energy security.

The SDS envisions a substantial reduction in oil demand by the 2030s, driven by the rise of electric vehicles (EVs) and alternative fuels in sectors like transport, shipping, and aviation. Advanced economies, particularly in Europe and the Asia-Pacific, are leading the way in EV adoption, with projections that over 90% of car sales will be electric by 2040. This shift, along with increased material efficiency and recycling, is expected to decrease oil demand in petrochemicals in developed economies.

For developing economies like India and China, oil demand is projected to grow until around 2025 and 2030, respectively, after which it will start to decline. Sub-Saharan Africa, while seeing increased oil consumption, will achieve universal energy access by 2030, maintaining low per capita oil use.

In terms of natural gas, the SDS emphasizes adapting production to shifting demand and addressing environmental concerns, particularly through significant reductions in methane emissions. This includes measures like electrifying liquefied natural gas (LNG) processes, increasing the use of Carbon Capture, Utilization, and Storage (CCUS), and repurposing gas grids for low-carbon energy distribution.

¹⁴ Climate Action Tracker, [Countries | Climate Action Tracker](#), (last visited on November 24th, 2023)

¹⁵ Saddler, B., & Verheem, R. (1996). *Oil and gas companies: Contributors to sustainability or barriers!*

¹⁶ Adedoyin OO, Olalekan RM, Olawale SH, et al: *A review of environmental, social and health impact assessment (Eshia) practice in Nigeria: a panacea for sustainable development and decision making*. 9 J. MOJ Public Health 81, 83 (2020)

For the oil and gas industry, the first step in the SDS is drastically reducing its environmental impact. This involves curbing methane emissions, eliminating flaring, and enhancing efficiency. The SDS stresses the importance of CO₂ pricing and policy interventions to drive change, alongside investments in low-carbon technologies such as hydrogen and biofuels.

To align with the SDS and meet global emissions reduction targets, key actions include accelerating the transition to electric vehicles, enhancing petrochemical efficiency, and repurposing gas grids for low-carbon energy. The oil and gas sector must focus on reducing emissions intensity by over 50% by 2040 and embrace CO₂ pricing and policy-driven interventions. Supporting research, developing nature-based solutions, and engaging actively in the electricity sector will also be essential in meeting ambitious emissions reduction goals.

In summary, achieving the SDS requires a concerted effort across the energy sector, including the oil and gas industry. With well-designed policies, industry support, and large-scale deployment of low-carbon technologies, we can pave the way for a sustainable energy future.



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1 Overview of Natural Gas Sector

1.1 A brief outline of your jurisdiction's natural gas sector, including a general description of: natural gas reserves; natural gas production including the extent to which production is associated or non-associated natural gas; import and export of natural gas, including liquefied natural gas ("LNG") liquefaction and export facilities, and/or receiving and re-gasification facilities ("LNG facilities"); natural gas pipeline transportation and distribution/transmission network; natural gas storage; and commodity sales and trading.

India's natural gas sector is governed by a complex legal and regulatory framework encompassing statutes, rules, and policies designed to manage exploration, production, transportation, storage, and trade. The Petroleum and Natural Gas Regulatory Board (**PNGRB**) serves as the central authority in regulating India's natural gas sector.

Natural gas reserves and production

The estimated reserves of natural gas as on April 1, 2022 was at 1149.46 Billion Cubic Metres, with 23.45% in the Eastern offshore and 29.34% in Western offshore.¹ India's natural gas reserves are primarily located offshore in the Krishna-Godavari Basin, Bombay High, and the Rajasthan Basin. Onshore reserves are located in Assam and Gujarat. As of fiscal year (**FY**) 2023–24, total natural gas production was 35,717 Million Standard Cubic Metres. India produces both associated and non-associated natural gas. There is no segregation of non-associated natural gas or associated gas figures. Major producers include public sector entities alongside private players.

Import and export of natural gas

India is heavily reliant on imports of liquified natural gas (**LNG**) to meet domestic demand, with key suppliers including Qatar and the United States. Dahej LNG Terminal (Gujarat) is one of the largest LNG import terminals, operated by Petronet LNG Ltd. It has a capacity of 17.5 million tonnes *per annum* (**MTPA**). Natural gas exports are minimal, as domestic production prioritises internal consumption.

Natural gas pipeline transportation and distribution

As of March, 2022, there is an authorised network of 34,135 km natural gas pipeline. Pipeline transportation of natural gas is a critical component of the energy infrastructure. The Indian government aims to create a nationwide gas grid to

connect major cities, industrial hubs and unreserved regions to a central pipeline network. The plan involves constructing thousands of kilometres of pipelines under the Pradhan Mantri Urja Ganga scheme and the National Gas Grid Initiative, which will improve gas access across rural and urban areas, promote cleaner energy, and boost industrial growth. Major existing pipelines include the Hazira-Vijaipur-Jagdishpur (**HVJ**) pipeline. Jagdishpur-Haldia and Bokaro-Dhamra Pipeline (**JHBDPL**) is an ongoing key project aimed at connecting the eastern part of India to the national gas grid. It will provide access to natural gas for industries and cities in Uttar Pradesh, Bihar, Jharkhand, West Bengal, and Odisha.

Natural gas storage

India's natural gas storage capacity is integrated with LNG terminals, ensuring minimal disruptions in supply.

1.2 To what extent are your jurisdiction's energy requirements met using natural gas (including LNG)?

The demand for energy is met mostly by fossil fuels with coal being the primary contributor (~60%), oil (~29%) and natural gas contributes only around 6% to the total energy demand. Balance (5%) is met through a mix of non-conventional sources.²

1.3 To what extent are your jurisdiction's natural gas requirements met through domestic natural gas production?

India imports around 50% of the natural gas requirements.³ In 2021, Qatar was the biggest supplier of natural gas to India, accounting for 42% of the total imports, with the USA at 16%.⁴ In June 2024, the USA was the largest exporter of LNG with 851,000 tonnes of supply overtaking Qatar.⁵

1.4 To what extent is your jurisdiction's natural gas production exported (pipeline or LNG)?

India does not export natural gas to any country.

2 Overview of Oil Sector

2.1 Please provide a brief outline of your jurisdiction's oil sector.

India is the third-largest energy and oil consumer in the world.⁶

A net importer of oil with 87.9% of its total oil requirement met through imports,⁷ International Energy Agency (**IEA**)

estimates in their 2023–2030 forecast period that India will account for more than one-third of global oil demand growth.

During FY 2023–24, India consumed 234.3 Million Metric Tonnes (MMT) of petroleum products.⁸

2.2 To what extent are your jurisdiction's energy requirements met using oil?

Oil contributes to 29% of the total energy demand.

2.3 To what extent are your jurisdiction's oil requirements met through domestic oil production?

As at April 2024, estimated recoverable reserves of crude oil in India stood at 671 MMT.

During FY 2023–24, domestic crude oil production was only 29.36 MMT, accounting for a modest ~12% oil requirement.⁹

The production has decreased in the last 10 years by 22%; however, consumption has increased by 47%.

2.4 To what extent is your jurisdiction's oil production exported?

Being a net-importer, Indian policy does not allow export of any crude oil produced.

3 Development of Oil and Natural Gas

3.1 Outline broadly the legal/statutory and organisational framework for the exploration and production ("development") of oil and natural gas reserves including: principal legislation; in whom the State's mineral rights to oil and natural gas are vested; Government authority or authorities responsible for the regulation of oil and natural gas development; and current major initiatives or policies of the Government (if any) in relation to oil and natural gas development.

The broad legal framework for the development of oil and natural gas reserves comprises of the Constitution of India, statutes and case laws.

India follows a federal structure. The Constitution of India establishes a division of legislative powers between the union and state governments. According to Entry 53 of List I in the Seventh Schedule of Constitution of India, the union government has exclusive legislative competence over "*the regulation and development of oil fields and mineral oil resources; petroleum and petroleum products; and other substances declared by Parliament to be dangerously inflammable*". State governments, however, can enact laws over certain natural resources that are imperative for extracting and doing other operations relating to oil and natural gas blocks namely: "Land"¹⁰; "Water"¹¹; and "Gas and Gas Work"¹².

The seminal legislation governing the oil and natural gas development industry in India is the Oilfields Regulation and Development Act, 1948 (ORD, 1948). However, much of the meat of the legal framework is in the Petroleum & Natural Gas Rules, 1959 (PNGR, 1959).¹³ PNGR, 1959 mandate the taking of a "license" for exploration and a "lease"¹⁴ for production. The Act is applicable to mineral oil,¹⁵ which includes only two categories of hydrocarbons viz. natural gas and oil. Further, the union government reserved the power to make rules for

altering the terms of the leases including rent, duration, area, etc.¹⁶ Any person who did not take such a licence or a lease suffers penal and/or pecuniary consequences.

The state's mineral rights to oil and natural gas are vested in the government. PNGR, 1959 mandate the taking of a "license" for exploration and a "lease" for production. This licence or lease is to be taken from the union government for offshore blocks and state government for onshore blocks.¹⁷ Similarly, the Territorial Waters, Continental Shelf, Exclusive Economic Zone and other Maritime Zones Act, 1976 provide for the grant of a licence by the government to explore and exploit the resources of the continental shelf and exclusive economic zone. By virtue of Article 297 of the Constitution of India, ownership of resources found within India's territorial waters, continental shelf, and exclusive economic zone is vested in the central government.

Under the Allocation of Business Rules, 1961,¹⁸ the chief government authority responsible for the regulation of oil and natural gas development in the country is the Ministry of Petroleum & Natural Gas, Government of India (MOP&NG). MOP&NG exercises regulatory oversight over the sector through its technical arm Directorate General of Hydrocarbons (DGH). DGH's objectives are to promote sound management of the oil and natural gas resources, with a balanced consideration for the environment, safety, technological and economic aspects of petroleum activity.

Current major initiatives and policies of the government include an amendment to overhaul ORD, 1948 and Hydrocarbon Exploration Licensing Policy (HELP)/Open Acreage Licensing (OAL) Policy.

Recently,¹⁹ the Oilfields (Regulation and Development) Amendment Bill, 2024 has been proposed in the union parliament which brings about notable changes to ORD, 1948 aimed at bringing policy stability. The Amendment is yet to be made into a law.

Key features of the amendment are:

- The most troublesome knot of applying twice to the government for a licence or lease has been deleted by adding a new definition of "petroleum lease",²⁰ which will be applicable for both exploration and development.
- Express inclusion of stabilisation provision for leases, which states that terms of leases will not be varied during their term to the disadvantage of investors. This will bring much comfort for investors.
- Augmentation to rule making powers of the union government:
 - to bring about alternative methods for dispute resolution in the context of leases including seats of such mechanism being both inside and outside India. This will bring much assurance and comfort to foreign investors;
 - for unitisation of oil and gas fields even across states so that reservoirs which extend to two different leases could be efficiently developed; and
 - to enhance, protect and conserve the environment.
- Penalties for contravention restricted to pecuniary consequences only, unlike penal provisions existing in the past. The penalties have, however, been increased from INR 1,000 to INR 25 lakhs. For the adjudication of offences, there will be a special authority not below the rank of Joint Secretary and appeals against their decision will go to Appellate Tribunal for Electricity (APTEL).

Further, the proposed rule-making powers also indicate that Government is planning to lay down a legislative framework

for comprehensive energy projects, which will include both oil fields as well as renewable sources.

HELP/OAL Policy is discussed in question 3.2 below.

3.2 How are the State's mineral rights to develop oil and natural gas reserves transferred to investors or companies ("participants") (e.g., licence, concession, service contract, contractual rights under Production Sharing Agreement?) and what is the legal status of those rights or interests under domestic law?

The state's mineral rights to develop oil and natural gas reserves are transferred to participants through an auction process.

HELP was launched by the central government in 2016, which is based on Revenue Sharing, replacing the previous cost recovery framework of Production Sharing. A key feature of HELP is the OAL mechanism, which allows investors to select exploration blocks based on their assessment of E&P data available through the National Data Repository (NDR). Investors can submit an Expression of Interest (EoI) at any time throughout the year, without waiting for a formal bidding round from the government. These blocks are then offered through a biannual formal bidding process.

The HELP bidding process is based on three key parameters: the quantum of work programme; revenue share; and originator incentive. The block is awarded to the bidder offering the highest commitment in terms of the work programme and the government's share of revenue. Additionally, an originator incentive is granted to the entity that identifies and carves out a block from the NDR and submits the EoI.

Rule 5 of PNGR, 1959 provides for an agreement between the government and the licensee or lessee containing additional terms and conditions with respect to the licence or lease. After the block is awarded to the successful bidding participant, the Revenue Sharing Contract is signed which contains additional terms related to term of exploration, work programme and budgets, sale of petroleum, dispute resolution, etc.

The legal status of the rights and interests to develop oil and natural gas reserves is both contractual and regulatory in nature. These rights take the form of a "license" during the exploration stage, which does not grant any interest in oil and gas. As the oil and gas fields progress towards development, the rights take the form of a "lease" granting the participants an interest to extract, utilise or sell hydrocarbons. The legal status is also considered regulatory because certain terms of these licences or leases are embodied in PNGR, 1959. Further, these rights are heavily regulated through the pervasive control of the government.

3.3 If different authorisations are issued in respect of different stages of development (e.g., exploration appraisal or production arrangements), please specify those authorisations and briefly summarise the most important (standard) terms (such as term/duration, scope of rights, expenditure obligations).

As stated in question 3.1 above, a "license" is issued for the exploration/appraisal stage and "lease" is issued for the development/production stage.

The standard terms are provided below:

Item	Licence	Lease
Duration	Category I Basin: ²¹ Three years for Onland/ Coal Bed Methane/ Shallow Water Blocks ²² and four years for Deep Water/Ultra Deep Water ²³ Blocks. Category II Basin ²⁴ & Category III Basin: ²⁵ Seven years.	20 years.
Scope	Exploration & Appraisal.	Development and Production.
Obligations	The Participant is required to complete the Committed Work Program, which comprises of exploratory survey and drilling work.	The Participant is required to commence commercial production, which comprises of exploratory survey and drilling work.
Default	In case of failure to complete the Committed Work Program, liquidated damages are payable in the range of USD 1 million to USD 12 million for Onland and Ultra Deep Water blocks respectively.	In case of failure to commence production, liquidated damages are payable as follows: Delay of (i) two years in case of Onland Blocks; (ii) three years in case of Shallow Water Blocks; or (iii) five years in case of Blocks falling in Deep Water/Ultra Deep Water areas from the date of grant of lease, one time fixed payment of USD 2 million for Onland Blocks and USD 10 million for Offshore Blocks and USD 2,000 per day over and above the fixed payment, until the date Commercial Production is commenced subject to a maximum delay of two years.

3.4 To what extent, if any, does the State have an ownership interest, or seek to participate, in the development of oil and natural gas reserves (whether as a matter of law or policy)?

The state has an economic interest in the development of oil and natural gas reserves, which it takes through royalty, revenue share, taxes, etc. However, all crude oil and natural gas produced has to be sold domestically in India. The state used to exercise deep and pervasive control over oil and natural gas development; however, this has been gradually removed by imposing economic sanctions on participants.

Until 1980, India had a nomination regime where oil and gas exploration rights were only given to state owned entities. In the 1980s, International Competitive Bidding (ICB) for oil and gas exploration blocks was introduced. However, significant fiscal incentives (such as right to acquire more interest in case of discovery) were provided to state-owned companies, creating a somewhat skewed playing field for private investors. The real turning point for India's hydrocarbon sector came in 1997 with the launch of the New Exploration Licensing Policy (NELP), which created a level playing field by offering private investors the same fiscal and contractual terms as those available to government companies.

Petroleum granting instruments such as Production Sharing Contract (**PSC**) and Revenue Sharing Contract (**RSC**) provide for constituting a “Management Committee” (**MC**), which consists of representatives from the state. In the pre NELP and NELP regimes, the MC was responsible for approving work programmes and budgets, field development plans, abandonment and site restoration, etc. To promote ease of doing business and attract investments, with the advent of HELP, the MC’s role has been gradually reduced to an observing and advisory body only.

3.5 How does the State derive value from oil and natural gas development (e.g., royalty, share of production, taxes)?

The state derives economic value from oil and natural gas development through royalty, production/revenue share, taxes and rentals. In addition, by placing an obligation to sell all domestically produced oil and natural gas in India, the state ensures energy security.

Production/revenue share

Under the PSC regime, cost recovery was allowed. This meant that participants can deduct costs of exploration to determine profit petroleum. The participants used to bid the profit share of the government, which formed the basis of winning blocks.

Under the RSC regime, the government takes a share of the “Revenue”. “Revenue” is all amounts that are accruing to the Contractor, net of any taxes and duties (except corporate income tax), and subtraction of royalty for that month.²⁶

The revenue share varies on the level of revenue achieved. The level of revenues are classified as “Low Revenue Point” (**LRP**) and “High Revenue Point” (**HRP**), which are expressed by the government when it is inviting offers for blocks.²⁷ The revenue share of government is a biddable range which is expressed as a percentage of total revenue at the LRP and HRP points.

In recent HELP rounds, the revenue is payable only for Category I basins. For other basins, the revenue share to the government is payable only if there is a revenue of more than USD 2.5 billion (windfall gain).

Royalty

Royalty rates range from 2% to 12%, depending on the basin category and shore status of blocks. Concessional royalty rates are provided in case there is early production.

Tax

Under the Income Tax Act, 1961, Corporate Income Tax at the rate of 30% is applicable. Allowances for expenditure incurred towards unsuccessful exploration is allowed.²⁸ In the case of production, 100% expenditure incurred towards exploration as a deduction can be made in the first year or amortised over a period of 10 years.

Surface rent

A miniscule deposit and yearly rent is payable to the government as licence or lease fees.²⁹

3.6 Are there any restrictions on the export of production?

As stated in question 2.4 above, being a net-importer, India does not export any crude oil. Petroleum granting instruments

such as PSCs/RSCs have an obligation to sell domestically produced crude oil in India until such time as the total availability of crude oil from all Petroleum production activities in India meets the total national demand.³⁰

3.7 Are there any currency exchange restrictions, or restrictions on the transfer of funds derived from production out of the jurisdiction?

In the oil and gas sector, there are no additional restrictions on fund remittances beyond the standard regulations applicable to other industries. The sector allows 100% Foreign Direct Investment (**FDI**) (through automatic route)³¹ under the Indian FDI policy, allowing foreign entities to operate without any sector-specific restrictions, subject to compliance with general tax and regulatory requirements.³² However, it is pertinent to note that companies based in nations bordering India face a stricter regulatory scrutiny, requiring heightened compliance awareness in the Indian FDI landscape.³³ Even in sectors where 100% FDI is permitted through the automatic route, such as oil and petroleum (except for Public Sector Undertaking, not under a strategic disinvestment scheme, where a 49% cap applies), entities from neighbouring countries are subject to the government approval route for any investment. This means that, regardless of the sectoral cap or the automatic nature of the route for other investors, these entities must obtain prior approval from the government before proceeding with any transaction.

3.8 What restrictions (if any) apply to the transfer or disposal of oil and natural gas development rights or interests?

The consent of DGH is required for transfer of oil and natural gas development interests. The consent is not unreasonably withheld; however, the government considers technical and financial capability of transferees before giving consent. Further, the government usually does not give such consent if there are sanctions or public policy considerations in respect of potential transferees.

For assignment to affiliates, there is no technical or financial assessment, however; parent company guarantees are required. An assignment or transfer shall not be made where the participating interest to be retained by the proposed assignor or the percentage interest of assignee shall be less than 10% of the total participating interest of all the participants, except where the government, on the recommendations of the MC may, in special circumstances, so permit.

Change of control is also subjected to the same procedure.

3.9 Are participants obliged to provide any security or guarantees in relation to oil and natural gas development?

Yes, participants are obliged to provide security and guarantees in relation to oil and natural gas development.

The participants must provide a bank guarantee to DGH for an amount equivalent to their participating interest share of liquidated damages for exploration.³⁴ If the block progresses towards development, the participants are required to submit Bank Guarantees, which may extend up to 5% of the total expenditure proposed for development. This quantum is dependent on the peculiarity of the blocks and should be seen from the documents available at auction.

In addition, the participants are also required to provide a parent company guarantee to DGH for assurance of

performance. In case there is no parent company, the participant itself must provide such guarantee.

There are standard formats of these guarantees annexed to the model contracts, which are provided at the start of a bidding round.

3.10 Can rights to develop oil and natural gas reserves granted to a participant be pledged for security, or booked for accounting purposes under domestic law?

Yes, rights to develop oil and natural gas reserves granted to a participant under the PSC/RSC can be pledged for security, or booked for accounting purposes under domestic law.³⁵

There are a few conditions which must be followed:

- The participant will remain liable for its obligations under the PSC/RSC.
- The pledge shall be subordinated to the rights of government under PSC and the rights of other consortium members under Joint Operating Agreements.
- Prior consent of government will be required for a list of lenders to be considered for creating such a pledge.

3.11 In addition to those rights/authorisations required to explore for and produce oil and natural gas, what other principal Government authorisations are required to develop oil and natural gas reserves (e.g., environmental, occupational health and safety) and from whom are these authorisations to be obtained?

In India, developing oil and natural gas reserves requires multiple government authorisations beyond those related to exploration and production. These include approvals concerning environmental protection, land acquisition, occupational health and safety, and other regulatory compliances.

The Environmental Protection Act, 1986, and the Environment Impact Assessment (EIA) Notification, 2006,³⁶ mandate obtaining prior Environmental Clearance (EC) for oil and gas projects. This involves conducting an EIA study and public consultation. The clearance is granted by the Ministry of Environment, Forest and Climate Change and State Environment Impact Assessment Authority. Under the Water (Prevention and Control of Pollution) Act, 1974,³⁷ and the Air (Prevention and Control of Pollution) Act, 1981,³⁸ developers must obtain "Consent to Establish" and "Consent to Operate" from the respective State Pollution Control Boards (SPCBs).

Projects must also comply with labour laws such as the Mines Act, 1952,³⁹ which regulates the health and safety of workers, overseen by the Directorate General of Mines Safety (DGMS). DGMS is a regulatory agency that regularly inspects mines to monitor safety, investigates accidents, dangerous occurrences, complaints, and emergency responses. It grants statutory permissions, exemptions, and relaxations.

If the project is located near coastal areas, clearance under the Coastal Regulation Zone Notification, 2019, issued under the Environmental Protection Act, 1986, is required from the Ministry of Environment, Forest and Climate Change or the respective state coastal zone management authority.

The Petroleum & Explosives Safety Organization (PESO) approval is also required for storing petroleum exceeding 2,500 litres.

Apart from the above, depending on the number of people employed in the extraction activity, labour compliances are also mandated.

3.12 Is there any legislation or framework relating to the abandonment or decommissioning of physical structures used in oil and natural gas development? If so, what are the principal features/requirements of the legislation?

Yes, there is a framework for site restoration and abandonment in the form of guidelines issued by DGH.⁴⁰

In addition to submission and seeking approval of DGH through MC, the guidelines postulate submission of the abandonment plan to Oil Industry Safety Directorate (OISD) for offshore production sites and DGMS for onshore production sites. In addition, the plan needs to be submitted to other authorities such as pollution control boards who may provide advisory comments on the plan. A high-level schedule of the abandonment plan needs to be submitted to DGH one year prior to the estimated cessation of production activities. The final plan needs to be submitted to OISD/DGMS no later than one year after cessation of production activities.

Under the Site Restoration Fund Scheme, 1999, the participants are required to constitute a fund which is to be maintained in the State Bank of India for funding the estimated decommissioning costs. The fund is to be opened immediately after first commercial production of oil/gas. An annual contribution is required to be made, which is based on the unit of production method, i.e. reserve of the field to production ratio. The fund contribution needs to be reviewed every three years by an independent third party. Withdrawals from the fund are permitted to pay for abandonment costs.

3.13 Is there any legislation or framework relating to gas storage? If so, what are the principal features/requirements of the legislation?

Yes, there are legislations in the Indian jurisdiction that govern the storage of gas.

The PNGRB Act establishes the PNGRB, which is responsible for authorising and regulating activities in the natural gas sector, including storage and transportation.⁴¹ Entities setting up storage facilities that exceed specific thresholds under the Act's various regulations are required to register with the PNGRB.⁴² Additionally, the Act empowers the central government to take over any storage site in the public interest during emergencies such as war or natural calamities.⁴³

There are other regulations, under this framework such as the Petroleum and Natural Gas Regulatory Board (Technical Standards and Specifications including Safety Standards for Liquefied Petroleum Gas (LPG) Storage, Handling and Bottling Facilities) Regulations, 2019, which outline detailed safety and engineering standards for LPG facilities, covering layout design, maintenance, fire protection, and emergency management.⁴⁴ Similarly, the Petroleum and Natural Gas Regulatory Board (Technical Standards and Specifications including Safety Standards for Petroleum Refineries and Gas Processing Plants) Regulations, 2023, provide guidelines for managing the storage, processing, and handling of hydrocarbons and hazardous substances within refineries and gas plants.

Additionally, there are standards set by the Bureau of Indian Standards that must be adhered to for the establishment of storage facilities. For instance, Codes of Practice for liquefied petroleum gas storage installations⁴⁵ lay down the code of practice for the installation of LPG cylinders, piping and equipment in commercial and industrial premises (including installations where storage tanks over 450 litres are used).⁴⁶

Considering the safety hazards associated with gas storage, India has framed comprehensive rules under the Indian

Explosives Act, 1884 for regulating storage of gas – Gas Cylinder Rules, 2016 (**GCR, 2016**) for low volumes⁴⁷ (up to 1,000 litres) and Static and Mobile Pressure Vessels (Unfired) Rules, 2016 (**SMPV, 2016**) for high volumes.⁴⁸

GCR, 2016 requires any person intending to fill and possess such cylinder to take approval from Chief Controller of Explosives, an authority designated under the Explosives Act, 1884. There are detailed specifications and technical standards provided in the rules for manufacturing cylinders.

SMPV 2016, regulates gas storage infrastructure for both pressure vessels as well as cryogenic vessels. Again, the rules provide taking approval from Chief Controller of Explosives. Special emphasis is given on minimum safety distance between vessels for storage of gas.⁴⁹

3.14 Are there any laws or regulations that deal specifically with the exploration and production of unconventional oil and gas resources? If so, what are their key features?

There are no special laws or regulations that deal specifically with exploration and production of unconventional oil and gas resources. The same regime of ORD, 1948 and the PNGR, 1959 is applicable to unconventional oil and gas resources.

With the advent of HELP, a single unified licence is applicable for prospecting both conventional and unconventional resources.

The recent amendment proposed to ORD, 1948, as discussed in question 3.1 above, also postulate an expansion to the definition of “mineral oil” to include coal bed methane and shale for legislative clarity.

3.15 What has been the impact, if any, of the “energy transition” on the oil and gas industry in your jurisdiction, and are there any policies or laws/regulations that require the oil and gas industry to decarbonise? Are there any policies or laws/regulations relating to the development of low-carbon hydrogen and its use in conjunction with or in place of natural gas, or the development of carbon capture and storage?

The worldwide shift toward energy transition for achieving net-zero emissions has significantly influenced India’s oil and gas industry. Working in furtherance of its nationally determined contribution to reducing carbon intensity by 45% by 2030 and achieving net-zero emissions by 2070,⁵⁰ the Indian government has introduced various policies and regulations to decarbonise the sectors with high emissions (oil and gas being one of them). These initiatives are also directed towards reducing dependency on non-renewable fuels and making India a net export in the sector.⁵¹

The National Green Hydrogen Mission is aimed at incentivising the production and use of green hydrogen.⁵² This initiative is expected to decarbonise the energy sector among others and reduce reliance on imported fossil fuels. It also focuses on developing domestic manufacturing capabilities, advancing technologies such as efficient fuel cells. Additionally, the Perform, Achieve, Trade Scheme (**PAT**) is a market-based regulatory tool to improve sector-wise energy efficiency. The energy sector was included in it in the year 2015.

SATAT Scheme is an umbrella scheme that unites various ministries and their initiatives under it.⁵³ This scheme integrates various ministries in a “whole of government” approach, targeting the production of compressed bio-gas

(**CBG**) from municipal, sugar industry, and agricultural waste. Complementary programmes like the National Bioenergy Programme and incentives under the Agriculture Mechanisation and Crop Residue Management Scheme contribute in a collaborative way to reduce emissions.⁵⁴

The Ethanol Blended Petrol (**EBP**) Programme, a significant step in this direction, promotes blending ethanol with petrol. Public Sector Oil Marketing Companies are incentivised to actively sell ethanol-blended petrol to reduce carbon emissions particularly in terms of vehicular emissions. Further, the Pradhan Mantri JI-VAN Yojana provides financial assistance to 2G bio-ethanol projects, enhancing biofuel production.⁵⁵

To synchronise the use of CBG and natural gas, guidelines for blending CBG with compressed natural gas (**CNG**) in city gas distribution networks have been introduced.⁵⁶ The National Biofuels Coordination Committee has also set a phase-wise mandatory blending target for CBG and CNG, beginning in the fiscal year 2025–2026.⁵⁷

4 Import / Export of Natural Gas (including LNG)

4.1 Outline any regulatory requirements, or specific terms, limitations or rules applying in respect of cross-border sales or deliveries of natural gas (including LNG).

India, as a net importer of natural gas, does not engage in cross-border sales.

Additionally, India currently lacks cross-border LNG pipelines.⁵⁸

5 Import / Export of Oil

5.1 Outline any regulatory requirements, or specific terms, limitations or rules applying in respect of cross-border sales or deliveries of oil and oil products.

The Foreign Trade (Development and Regulation) Act, 1992 serves as the foundational statute regulating all exports from India. Under Section 5 of this Act, the export policy is notified by the central government via the Foreign Trade Policy (**FTP**). The current FTP (2023–28) outlines the conditions and restrictions for exporting petroleum products, which may include quotas, quality specifications, or embargoes based on international agreements or national interests. In addition, sectoral prohibitions for export, such as refined petroleum products, cannot be exported to the Democratic People’s Republic of Korea. Exporters are required to adhere to the directives of the Directorate General of Foreign Trade (**DGFT**) and obtain mandatory licences or permissions under the FTP.

Further, the Customs Act, 1962 imposes duties and levies on exported petroleum products unless exempted under the FTP or specific exemptions provided in customs notifications.

6 Transportation

6.1 Outline broadly the ownership, organisational and regulatory framework in relation to transportation pipelines and associated infrastructure (such as natural gas processing and storage facilities).

Under the PNGRB Act, the PNGRB has been established to regulate the refining, processing, storage, transportation,

distribution, marketing, and sale of petroleum, petroleum products, and natural gas. The PNGRB Act lays down that any entity desirous of marketing, establishing, or operating natural gas or petroleum terminals or any entity interested in establishing storage facilities for petroleum, petroleum products, or natural gas is required to take prior authorisation.

The PNGRB (Authorising Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations, 2008 establishes a structured bidding process for entities seeking authorisation to develop or expand natural gas pipelines.⁵⁹ It lays down an elaborate process for laying, building, operating and expanding⁶⁰ natural gas pipelines. This includes, *inter alia*, initiation of proposal through expression of interest or *suo-motu* (by the board) routes,⁶¹ criteria for selection of entity for expression of interest route,⁶² and bidding criteria,⁶³ etc.

In addition to the above, the PNGRB has introduced separate regulations governing tariff, marketing and technical specification.

Additionally, the Petroleum Act, 1934, in conjunction with the Petroleum Rules, 2002,⁶⁴ governs the import, transport, and storage of petroleum products.⁶⁵ Depending on the hydrocarbon composition, natural gas and LNG may also fall under the ambit of this legislation, thus making it applicable to the broader transportation pipeline framework.⁶⁶

6.2 What governmental authorisations (including any applicable environmental authorisations) are required to construct and operate oil and natural gas transportation pipelines and associated infrastructure?

The PNGRB is authorised under the PNGRB to authorise entities to lay, build, operate or expand a common carrier or contract carrier.

Apart from the authorisation under the PNGRB Act, to develop a natural gas pipeline, the relevant entity should obtain multiple clearances and land rights based on the route that the pipeline is taking. The authorisation under the PNGRB Act does not enable the entity to obtain the other related clearances to develop the pipeline in any preferential or assured manner. The relevant entity must separately undertake obtaining environmental clearances, forest land use clearances, clearances to cross highways, railways, public roads, municipal areas, acquire right-of-use through the specific process stipulated under the Petroleum and Minerals, Pipelines (Acquisition of Right of User in Land) Act 1962.

Under the Office Memorandum, dated September 6, 2018, issued by the Ministry of Environment, Forest and Climate Change, Government of India, only oil and gas transportation pipeline passing through national parks, sanctuaries, coral reefs, ecologically sensitive areas including LNG terminal require prior environmental clearance from the concerned regulatory authority.

6.3 In general, how does an entity obtain the necessary land (or other) rights to construct oil and natural gas transportation pipelines or associated infrastructure? Do Government authorities have any powers of compulsory acquisition to facilitate land access?

The relevant entity can acquire right-of-use through the specific process stipulated under the Petroleum and Minerals, Pipelines (Acquisition of Right of User in Land) Act 1962. An entity can either negotiate privately with landowners or seek to compulsorily acquire the right of use in landfalling along

its route using the provisions of the Petroleum and Minerals, Pipelines (Acquisition of Right of User in Land) Act 1962, which provides a specific framework by which a pipeline entity can compulsorily acquire right of use in landfalling on the route of its pipeline through a process assisted by the government administration. The process involves the evaluation of right-of-use and payment of compensation.

6.4 How is access to oil and natural gas transportation pipelines and associated infrastructure organised?

Presently, around 75% of all natural gas pipelines operating in India are owned and controlled by the state-owned gas company, GAIL. There is also no assurance of obtaining interconnection with other natural gas pipelines and separately negotiated agreements for any interconnection needs to be undertaken. However India has the concept of common carrier and contract carrier, which is provided in detail in question 6.6.

6.5 To what degree are oil and natural gas transportation pipelines integrated or interconnected, and how is co-operation between different transportation systems established and regulated?

The Indian government has been actively developing a National Gas Grid to ensure the availability and equitable distribution of natural gas nationwide.

The Petroleum and Natural Gas Regulatory Board has promulgated regulations for declaring natural gas and petroleum product pipelines as common carrier along with detailed access codes. The regulations obligate pipeline owning entities to declare available capacity, entry-exit points, calorific value of gas which may be transported, etc. provided in detail in question 6.6.

6.6 Outline any third-party access regime/rights in respect of oil and natural gas transportation and associated infrastructure. For example, can the regulator or a new customer wishing to transport oil or natural gas compel or require the operator/owner of an oil or natural gas transportation pipeline or associated infrastructure to grant capacity or expand its facilities in order to accommodate the new customer? If so, how are the costs (including costs of interconnection, capacity reservation or facility expansions) allocated?

The Petroleum and Natural Gas Regulatory Board (Guiding Principles for Declaring or Authorizing Natural Gas Pipeline as Common Carrier or Contract Carrier) Regulations, 2009, Petroleum and Natural Gas Regulatory Board (Guiding Principles for Declaring or Authorizing Petroleum and Petroleum Products Pipelines as Common Carrier or Contract Carrier) Regulations, 2012, along with access code, permit the interconnection of common or contract carrier pipelines, provided there is sufficient capacity in the receiving pipeline and such interconnection is both technically and operationally feasible.

The extra capacity for use as common carrier by any third party on open access and non-discriminatory basis is allocated on first come first serve basis ensuring fair utilisation of the pipeline infrastructure. These regulations mandate that the operating entity shall offer the extra capacity in the pipeline to third parties on a common carrier or contract carrier basis.

However, such access is only contingent on the existence of extra capacity in the pipeline and the customer cannot compel the operator unless there is surplus unutilised capacity.

The PNGRB also has the power to declare an existing pipeline as common carrier. Detailed regulations have been laid out by the PNGRB in this regard.

For costs associated with such access, the third party would have to bear the transmission tariff set by the PNGRB. Such tariff is fixed on a zonal postalised basis and is recovered by the operating entity from all customers on a non-discriminatory basis, without premium or discount.

6.7 Are parties free to agree the terms upon which oil or natural gas is to be transported or are the terms (including costs/tariffs which may be charged) regulated?

In India, the transportation of petroleum product and natural gas is subject to regulatory oversight to ensure fair access and reasonable tariffs. While parties can negotiate certain terms of transportation agreements, key aspects, particularly tariffs, are regulated by the PNGRB.

The Petroleum and Natural Gas Regulatory Board (Determination of Natural Gas Pipeline Tariff) Regulations, 2008 (**Tariff Regulations**) regulates the determination of tariff for natural gas pipelines. The PNGRB, in March 2023, introduced the Unified Tariff System by making changes to the Tariff Regulations to promote the "One Nation, One Grid, One Tariff" initiative.

Similarly, determination of tariff for petroleum product pipelines is regulated under the PNGRB (Determination of Petroleum and Petroleum Products Pipeline Transportation Tariff) Regulations, 2024. With the latest amendments, introduced in July 2024, an independent criteria is relied on for setting up of pipeline tariffs.

7 Gas Transmission / Distribution

7.1 Outline broadly the ownership, organisational and regulatory framework in relation to the natural gas transmission/distribution network.

The regulatory framework for natural gas transmission and distribution in India is governed primarily by the PNGRB. The PNGRB grants licences to city gas distribution (CGD) networks in geographical areas (**GAs**) through competitive bidding, considering criteria like the number of households, network coverage, and proposed investments. Authorised pipelines must adhere to the common carrier or contract carrier principle.

7.2 What governmental authorisations (including any applicable environmental authorisations) are required to operate a distribution network?

The PNGRB is empowered to authorise entities to lay, build, operate or expand a city or local natural gas distribution network.

7.3 How is access to the natural gas distribution network organised?

Access to natural gas and petroleum pipelines outside local or city gas distribution is covered in response to question 6.4.

7.4 Can the regulator require a distributor to grant capacity or expand its system in order to accommodate new customers?

The authorised entity must meet the minimum annual work programme in the geographical area to develop or expand the gas distribution network, which comprises of the number of domestic piped natural gas connections which an entity has committed to install. During the exclusivity period of authorised entity, the PNGRB (Exclusivity for City or Local Natural Gas Distribution Network) Regulations, 2008 provide an obligation on the authorised entity to add additional domestic piped natural gas connection on demand to a domestic consumer for cooking purposes if such consumer is within a distance of 25 metres of the metering unit at the consumer's end until the tap-off in the pipeline. The authorised entity also has the right to provide a dedicated pipeline for any customer.

7.5 What fees are charged for accessing the distribution network, and are these fees regulated?

The tariff quoted by the winning CGD entity is charged to the customers. The authorised entity may also take an interest-free refundable security deposit from domestic customer towards security of the equipment and facilities, including the labour cost of installation towards last mile connectivity. The Petroleum and Natural Gas Regulatory Board (Determination of Transportation Rate for CGD and Transportation Rate for CNG) Regulations, 2020 regulates the tariff for the CGD entity after its exclusivity period is over and the same has been declared as a common carrier.

7.6 Are there any restrictions or limitations in relation to acquiring an interest in a gas utility, or the transfer of assets forming part of the distribution network (whether directly or indirectly)?

Under the PNGRB (Authorizing Entities to Lay, Build, Operate or Expand City or Local Natural Gas Distribution Networks) Regulations, 2008, an entity is not permitted to renunciate its authorisation by way of sale, assignment, transfer or surrender to any person or entity for a certain period from the date of its issue.

8 Natural Gas Trading

8.1 Outline broadly the ownership, organisational and regulatory framework in relation to natural gas trading. Please include details of current major initiatives or policies of the Government or regulator (if any) relating to natural gas trading.

India has established a transparent market for natural gas trading through the Indian Gas Exchange (**IGX**). The PNGRB (Gas Exchange) Regulations, 2020, regulate the functioning of trading platforms, setting standards for registration, price discovery, and dispute resolution. No single entity (other than a member of the exchange) can hold more than 25% of the paid-up equity share capital of a gas exchange after five years from authorisation. Gas exchanges must establish a well-defined board of directors, including: independent directors to ensure neutrality and promote investor confidence. The

structure is designed to align with global best practices for exchange governance. Membership is divided into: trading members; clearing members; trading-cum-clearing members; and proprietary members.

Entities intending to establish a gas exchange must obtain authorisation from the PNGRB.

8.2 What range of natural gas commodities can be traded? For example, can only “bundled” products (i.e., the natural gas commodity and the distribution thereof) be traded?

The PNGRB regulates the processing, storage, distribution, marketing, and sale of petroleum products and natural gas. This act defines “natural gas”⁶⁷ as gas obtained from bore-holes, and includes, *inter alia*, liquified gas, compressed natural gas, gas imported through transnational pipelines, and natural gas recovered from gas hydrates, but excludes helium gas. Since the Gas Exchange Regulations⁶⁸ regulate the establishment, operation and other related matters connected to gas exchange in India, the range of natural gas commodities that can be traded would have to be in accordance with these regulations. The Gas Exchange Regulations prescribe the same meaning to natural gas as has been awarded under the PNGRB Act. Thus, natural gas commodities that fall within the above definition are permitted to be traded.

It is pertinent to mention that the PNGRB also has the power to extend the applicability of the Gas Regulations for derivative, forward and future contracts with respect to natural gas or liquefied natural gas.⁶⁹

9 Liquefied Natural Gas

9.1 Outline broadly the ownership, organisational and regulatory framework in relation to LNG facilities.

LNG Terminals are owned and operated by both private and government entities.

In 2012, the PNGRB (Eligibility Conditions for Registering of Liquefaction of Natural Gas Terminals) Rules, 2012 (**PNGRB LNG Rules**) were promulgated. Notably, these rules did not mandate registration but mandated providing common carrier capacity.

Now the PNGRB has come up with the new Draft PNGRB (Registration for Establishing & Operating LNG Terminals) Regulations, 2024, which mandate registration of LNG Terminals.

Currently, India has seven operational LNG terminals, all onshore, with an operational regasification capacity of around 170 MMSCMD.⁷⁰

9.2 What governmental authorisations are required to construct and operate LNG facilities?

Under the PNGRB LNG Rules, an entity proposing to develop or operate an LNG terminal is required to make an application before the PNGRB and is required to comply with the technical standards and specifications including safety standards prescribed by the PNGRB.

The PNGRB (Technical Standards and Specifications including Safety Standards for Liquefied Natural Gas Facilities) Regulations, 2018 prescribe: (a) lay down minimum requirements of layout within the plant boundary for unloading or loading, storage, regasification, transfer and handling and tank truck loading facilities for LNG facilities; (b) covers

safety in design and operational aspects of process systems, storage tanks, regasification facilities, ship shore interlock, berthing conditions for the ship, receiving facilities including jetty and port; and (c) covers engineering considerations in design, operations, maintenance, inspection and installations including fire protection and safety systems.⁷¹

For pipeline facilities related to LNG, please refer to section 6 above.

In addition to these regulations, entities must obtain necessary environmental clearances and comply with other statutory requirements as mandated by relevant authorities. For example, the Ministry of Shipping also has several technical requirements for establishing a Floating Storage Re-gasification Unit at major ports.

9.3 Is there any regulation of the price or terms of service in the LNG sector?

Pricing framework

The Natural Gas Allocation Policy (2014) administers allocation of gas from domestic fields to priority sectors like fertilisers, power, and city gas distribution.

Non-APM (Administered Price Mechanism) gas is sold under market-determined prices. The Unified Tariff System (2023) was introduced to simplify tariff structures by integrating multiple pipeline tariffs into a single zone-based tariff.

9.4 Outline any third-party access regime/rights in respect of LNG facilities.

Third-party access to LNG facilities is proposed to be regulated when Draft PNGRB (Registration for Establishing & Operating LNG Terminals) Regulations, 2024 is brought into effect.

10 Downstream Oil

10.1 Outline broadly the regulatory framework in relation to the downstream oil sector.

The downstream oil sector in India is heavily regulated since oil products like Motor Spirit (**MS**) Petrol, High Speed Diesel (**HSD**), LPG, etc. are considered as “Essential Commodities” under the Essential Commodities Act, 1955. Under the Act, the central government has the power to regulate the production, supply and distribution of these by issuing “control orders”. In this regard, the central government has issued the Petroleum Products (Maintenance of Production, Storage and Supply) Order, 1999. Apart from powers to regulate production, supply and distribution, the government also has the power to order takeover of any petrol pump (even without an opportunity of hearing) for ensuring uninterrupted supply.

The Motor Spirit and High-Speed Diesel (Regulation of Supply, Distribution and Prevention of Malpractices) Order, 2005 issued by the central government under the Essential Commodities Act, 1955 provides for punitive action against malpractices such as adulteration. The public sector Oil Marketing Companies (**OMCs**) have also formulated the Marketing Discipline Guidelines, which helped to maintain discipline in the operation of retail network and provide high customer service standards. These guidelines are required to be followed by dealers who have been appointed by OMCs to distribute petroleum products. In the event of an infraction such as short delivery, adulteration, overcharging, etc. the

OMCs take action against dealerships which includes penalty, suspension and termination.

10.2 Outline broadly the ownership, organisation and regulatory framework in relation to oil trading.

Crude Oil forms a Derivative Commodity under the Securities Contract Regulation Act, 1956. Crude oil is traded on the Multi Commodity Exchange also known as MCX. The Securities and Exchange Board of India regulates crude oil trading.

11 Competition

11.1 Which governmental authority or authorities are responsible for the regulation of competition aspects, or anti-competitive practices, in the oil and natural gas sector?

The Competition Act 2002 (**Competition Act**), read with the Competition Commission of India (Procedure in Regard to the Transaction of the business relating to Combinations), 2011 (**Combination Regulations**), governs the regulation of competition aspects or anti-competitive practices in India. The Competition Commission of India (**CCI**) is the sole statutory authority empowered to regulate the competition aspects or anti-competitive practices in India.

11.2 To what criteria does the regulator have regard in determining whether conduct is anti-competitive?

The Competition Act regulates two kinds of agreements: (a) anti-competitive agreements between/amongst competitors i.e., horizontal agreements; and (b) anti-competitive agreements between persons at different stages or levels of the production chain i.e., vertical agreements. Under the Competition Act, certain type of horizontal agreements are presumed to cause an appreciable adverse effect on competition in India (**AAEC**). The presumption does not mean that all alleged agreements are necessarily anti-competitive; the parties entering into such an agreement are required to provide evidence that the agreement does not result in an AAEC and refute the presumption. To the contrary, this presumption does not apply to vertical agreements. Vertical agreements are permitted unless it is established that they cause, or are likely to cause, an AAEC within India. The Competition Act provides an exhaustive list of horizontal agreements that are presumed to cause an AAEC in India, as well as an inclusive list of vertical agreements that may be prohibited depending upon their effect on conditions of competition within India. The CCI has the power to determine any conduct as anti-competitive irrespective of whether or not any merger or amalgamation is occurring or any thresholds.

11.3 What power or authority does the regulator have to preclude or take action in relation to anti-competitive practices?

The CCI has the power to inquire into any alleged contravention of the provisions relating to anti-competitive activities either on its own motion or on: (a) receipt of any information from any person, consumer or their association or trade association; or (b) a reference made to it by the Central Government or a State Government or a statutory authority.

If the CCI is of the opinion that there exists a *prima facie* case, it shall direct the relevant officer to cause an investigation to be made into the matter and, upon satisfaction: (a) direct any person involved in such agreement to discontinue and not to re-enter such agreement; (b) impose such penalty, as it may deem fit which shall be not more than 10% of the average of the turnover or income, as the case may be, for the last three preceding financial years, upon such person or enterprise which is a party to such agreement; (c) direct that the agreements shall stand modified to the extent and in the manner as may be specified in the order by the CCI; (d) direct the person concerned to abide by such other orders as the CCI may pass and comply with the directions, including payment of costs, if any; and (e) pass such other order or issue such directions as it may deem fit.

11.4 Does the regulator (or any other Government authority) have the power to approve/disapprove mergers or other changes in control over businesses in the oil and natural gas sector, or proposed acquisitions of development assets, transportation or associated infrastructure or distribution assets? If so, what criteria and procedures are applied? How long does it typically take to obtain a decision approving or disapproving the transaction?

In September 2024, India notified several amendments to the Competition Act, 2023, expected to have a major impact on both domestic and international corporate mergers. Post the amendment, the CCI requires companies with significant operations in India to notify the regulator of mergers and acquisitions and amalgamations (collectively called “**combinations**”) that exceed INR 20 billion. This follows the Ministry of Corporate Affairs setting this threshold under the Competition (Amendment) Act, 2023, with provisions to define “substantial business operations”. The latest amendments have shortened the merger review timeline by the CCI. The amendment specifies that M&A deals worth over INR 20 billion now require approval from the CCI, provided the target company has significant business operations in India (for example, turnover exceeding INR 5 billion or 10% of global metrics). This amendment aligns India's regulatory framework with global standards. This threshold ensures high-value deals are reviewed to prevent anti-competitive practices. Earlier, the CCI had 30 working days to form a *prima facie* view on a notified transaction, which has now been reduced to 15 days. Additionally, the overall review period has been reduced from 210 to 150 calendar days.

12 Foreign Investment and International Obligations

12.1 Are there any special requirements or limitations on acquisitions of interests in the natural gas sector (whether development, transportation or associated infrastructure, distribution or other) by foreign companies?

There are no investment caps on foreign companies in acquisition of exploration activities of oil and natural gas fields, infrastructure related to marketing of petroleum products and natural gas, marketing of natural gas and petroleum products, petroleum product pipelines, natural gas/pipelines, LNG regasification infrastructure and petroleum refining in the private sector.

12.2 To what extent is regulatory policy in respect of the oil and natural gas sector influenced or affected by international treaties or other multinational arrangements?

While India is not a signatory to any international treaty specific to the oil and natural gas sector, it is, however, a signatory to ancillary treaties which may impact the oil and natural gas sector in India. For instance, India has signed several mobile offshore units (**MoUs**) with other countries for promoting the development of oil and natural gas. India has also signed several Bilateral Investment Treaties (**BITs**) with several countries for protection of Investments. India is also a signatory to Indo-Pacific Economic Framework for Prosperity Agreement Relating to Supply Chain Resilience for promoting and protecting crude oil and natural gas supply chains.

While not legally binding, the IEA recommendations and analysis influence global energy policy trends. India engages with the IEA and considers its policy recommendations when formulating its own energy policies.⁷²

The International Cooperation Division of the Ministry of Petroleum and Natural Gas encourages India to collaborate under various frameworks like G20, QUAD, Shanghai Cooperation Organisation (**SCO**), Association of South-East Asian Nations (**ASEAN**), etc.⁷³

India has also entered into various Free Trade Agreements (**FTAs**) that influence trade in the Energy Sector, such as ASEAN-Indian Free Trade Area (**AIFTA**), which undertakes economic cooperation activities in various sectors such as services including mining and energy.⁷⁴ Moreover, it is pertinent to mention that India is currently involved in FTA negotiations in trade services with the United Kingdom, European Union, Oman, Peru and Sri Lanka.⁷⁵

13 Dispute Resolution

13.1 Provide a brief overview of compulsory dispute resolution procedures (statutory or otherwise) applying to the oil and natural gas sector (if any), including procedures applying in the context of disputes between the applicable Government authority/regulator and: participants in relation to oil and natural gas development; transportation pipeline and associated infrastructure owners or users in relation to the transportation, processing or storage of natural gas; downstream oil infrastructure owners or users; and distribution network owners or users in relation to the distribution/transmission of natural gas.

Dispute resolution in the oil and gas sector comprises of contractual as well as statutory mechanisms.

Oil and natural gas development

Any dispute arising under the PSCs/RSCs is typically resolved by arbitration on an *ad-hoc* basis (non-institutional). In case the participant is a government entity, there is a special dispute resolution mechanism under the Department of Public Enterprises, Government of India. This procedure involves an arbitration panel of government representatives to resolve the dispute. The formal Arbitration & Conciliation Act, 1996 is not applicable to such procedure and a separate guideline governs this procedure.⁷⁶

Transportation processing or storage of natural gas, downstream oil infrastructure owners or users and distribution network owners

Under the Petroleum & Natural Gas Regulatory Board Act, 2006, the Petroleum and PNGRB has the power to decide disputes relating to refining, processing, storage, transportation, distribution, marketing and sale of petroleum, petroleum products and natural gas unless the parties have agreed for arbitration.⁷⁷ Any person who is aggrieved by the order of the PNGRB has to appeal to APTEL constituted under the Electricity Act, 2003.⁷⁸ In case the appeal is unsuccessful, a final right of appeal is provided to the Supreme Court of India.⁷⁹

13.2 Is your jurisdiction a signatory to, and has it duly ratified into domestic legislation: the New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards; and/or the Convention on the Settlement of Investment Disputes between States and Nationals of Other States ("ICSID")?

India signed the New York Convention on the Recognition and Enforcement of Foreign Arbitral Awards on June 10, 1958 and ratified the same in domestic legislation by enacting the Foreign Awards (Recognition and Enforcement) Act 1961 (**1961 Act**). The 1961 Act applied to all "arbitral awards" not considered to be domestic made on or after October 11, 1960. Thereafter, the Arbitration and Conciliation Act, 1996 (**A&C Act**) repealed the 1961 Act. The A&C Act now governs the enforcement of foreign arbitrations in India.

India is not a party to ICSID.

13.3 Is there any special difficulty (whether as a matter of law or practice) in litigating, or seeking to enforce judgments or awards, against Government authorities or State organs (including any immunity)?

There is no special difficulty (whether as a matter of law or practice) in litigating, or seeking to enforce judgments or awards, against government authorities or state organs. There is no sovereign immunity in India and infact the Constitution of India⁸⁰ postulates that both central and state governments can be sued. The Supreme Court of India has held that liability of government under its contracts is akin to a private party.

13.4 Have there been instances in the oil and natural gas sector when foreign corporations have successfully obtained judgments or awards against Government authorities or State organs pursuant to litigation before domestic courts?

There have been challenges to arbitration award which are against the government. Most notable in the sector is the "Gas Migration" case which involved a consortium of foreign companies. In this case it was alleged by the government that the consortium had no right to take gas which migrated from an adjacent block. The arbitral tribunal was held in favour of the consortium; however, the award was challenged by the government. The challenge was dismissed by the Delhi High Court and the award was upheld.⁸¹

There have also been challenges to operation of PSCs, notably extensions, profit share etc. in which foreign corporations have obtained favourable results.

14 Updates

14.1 Have there been any new regulatory or policy initiatives in your jurisdiction directly in response to the continuing global concerns around higher oil and gas prices and energy security (such as price caps, subsidies or a new focus on local sources of energy)?

Import of crude oil in India takes place at international prices, which is issued daily by Platts assessment of benchmark “Brent” for sweet crude and average of “Dubai” and “Oman” for sour crude.

Domestically produced crude is being sold in the market at international prices corresponding to the crude assay. The government has recently⁸² approved revised guidelines for pricing of domestically produced natural gas, which has led to a decrease in prices of PNG for households and CNG.

The Indian government has been continuing to provide subsidies on Public Distribution Kerosene and LPG since 2002.

14.2 Please provide, in no more than 300 words, a summary of any new cases, trends and developments in Oil and Gas Regulation Law in your jurisdiction (other than anything already discussed above).

With twin objectives of ensuring policy stability and promoting ease of business, the amendments to ORD, 1948 intends to bring about a landmark change for the upstream industry in India. The oversight by the PNGRB over the pipeline sector is becoming more tight with several amendments to regulations and amendments for natural gas and CGD networks.

Endnotes

- 1 https://mospi.gov.in/sites/default/files/publication_reports/EnergyStatistics_India_publication_2024N.pdf
- 2 NITI Ayog data.
- 3 <https://www.financialexpress.com/business/industry-natural-gas-imports-surge-18-per-cent-to-7-7-billion-in-h1-3645789/>
- 4 <https://www.statista.com/statistics/1237488/lng-import-share-india-by-country/>
- 5 <https://www.thehindubusinessline.com/economy/indias-lng-imports-during-may-july-2024-at-4-year-high/article68562836.ece>
- 6 India Brand Equity Foundation report August, 2024.
- 7 International Energy Agency data.
- 8 Ministry of Petroleum & Natural Gas – Annual Report 2023–24.
- 9 *Ibid.*
- 10 The Constitution of India, Seventh Schedule, Entry 18, List II, ‘Land, that is to say, rights in or over land, land tenures including the relation of landlord and tenant, and the collection of rents; transfer and alienation of agricultural land; land improvement and agricultural loans; colonization’.
- 11 The Constitution of India, Seventh Schedule, Entry 17-List II, ‘Water, that is to say, water supplies, irrigation and canals, drainage and embankments, water storage and water-power...’
- 12 The Constitution of India, Seventh Schedule, Entry 17-List II, ‘Gas and Gas Work’.
- 13 Framed under Section 5 & Section 6 of the Oilfields Regulation and Development Act, 1948.
- 14 “Mining lease” means a lease granted for the purpose of searching for, winning, working, getting, making merchantable, carrying away

or disposing of mineral oils or for purposes connected therewith, and includes an exploring or a prospecting licence.

- 15 Includes natural gas and petroleum.
- 16 Section 5(1) – The central government may, by notification in the *Official Gazette*, make rules for regulating the grant of mining leases or for prohibiting the grant of such leases in respect of any mineral oil or in any area.

(2) – In particular, and without prejudice to the generality of the foregoing power, such rules may provide for all or any of the following matters, namely: (a) the manner in which, the mineral oils or areas in respect of which and the persons by whom, applications for mining lease may be made and the fees to be paid on any such applications; (b) the authority by which, the terms on which, and the conditions subject to which, mining leases may be granted; (c) the maximum or minimum area and the period for which any mining lease may be granted, and the terms on which leases in respect of contiguous areas may be amalgamated; and (d) the fixing of the maximum and minimum rent payable by a lessee, whether the mine is worked or not.
- 17 Section 5(1) & Section 5(2).
- 18 Rule 3 read with Schedule 2.
- 19 August 5, 2024.
- 20 “Petroleum lease” means a lease granted on or after the commencement of the Oilfields (Regulation and Development) Amendment Act, 2024, for the purpose of prospecting, exploration, development, production, making merchantable, carrying away or disposing of mineral oils or for purposes connected therewith, and includes a mining lease granted before the commencement of the said Act.
- 21 Basins, which have reserves and already producing.
- 22 Up to 400 metres.
- 23 Beyond 400 metres.
- 24 Basins, which have contingent resources pending commercial production.
- 25 Basins, which have prospective resources awaiting discovery.
- 26 Model Revenue Sharing Contract, Open Acreage Licensing Policy – IX, 2024.
- 27 Notice Inviting Offer, OALP – IX 2024.
- 28 S. 42 of Income Tax Act, 1961.
- 29 S.11 – S.13 Petroleum and Natural Gas Rules, 1959.
- 30 Model PSCs in NELP, OALP & DSF Rounds.
- 31 5.2.4, Ch. 5, Department for Promotion of Industry and Internal Trade, Ministry of Commerce and Industry, Consolidated FDI Policy (Effective October 15, 2020).
- 32 Section 5 and Section 6, The Foreign Exchange Management Act, 1999; The Companies Act, 2013; The Foreign Exchange Management (Non-Debt Instruments) Rules, 2019; The Income-tax Act, 1961.
- 33 Department for Promotion of Industry and Internal Trade, Ministry of Commerce and Industry, Press Note No. 3 (2020 Series) (Issued on March 17, 2020).
- 34 *Ibid.*
- 35 Inclusion of express causes in MRSC OALP IX and NELP PSCs.
- 36 Section 3(1) and Section 3(2)(v), Environment (Protection) Act, 1986; Rule 5(3)(d), Environment (Protection) Rules, 1986 and notification number S.O. 60 (E) dated the 27 January, 1994.
- 37 Section 25.
- 38 Section 21.
- 39 Chapters 5, 6 and 7.